

Space-time fractional stochastic heat equation with spatially inhomogeneous white noise

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April 3, 2021

Abstract

In this paper, we consider the space-time fractional stochastic heat equation

$$\partial_t^\beta u(t, x) = -\nu(-\Delta)^{\frac{\alpha}{2}} u(t, x) + I_t^{1-\beta} \left(\lambda \sigma(u(t, x)) \dot{F}_\rho(t, x) \right), \quad 0 \leq t \leq T, x \in \mathbb{R},$$

where $T > 0$, $\beta \in (0, 1)$, $\nu > 0$, $\alpha \in (1, 2]$, $\lambda > 0$ is the noise level and $\dot{F}_\rho(t, x)$ denotes the spatially inhomogeneous white noise. Under some mild assumptions, we study the moment bounds for the solution. We show that the p -th ($p \geq 2$) moment grows at most exponentially with time and the second moment grows at least exponentially in time. Along the way, the asymptotic behaviour of the solution with respect to the parameter λ is also studied.

2010 AMS Classification Numbers: 60H15, 60G15, 60G17.

Key words: Space-time fractional stochastic heat equation; Spatially inhomogeneous white noise; Moment bounds; Intermittency.

1 Introduction

In this work we study the following nonlinear space-time fractional stochastic heat equation (STF-SHE for short)

$$\begin{cases} \partial_t^\beta u(t, x) = -\nu(-\Delta)^{\frac{\alpha}{2}} u(t, x) + I_t^{1-\beta} \left(\lambda \sigma(u(t, x)) \dot{F}_\rho(t, x) \right), & 0 \leq t \leq T, x \in \mathbb{R}, \\ u(0, x) = u_0(x), & x \in \mathbb{R}, \end{cases} \quad (1)$$

*Partially supported by Humanities and Social Sciences Foundation of the Ministry of Education of China (No. 18YJCZH101), Major Research Plan of Natural Science Foundation of the Jiangsu Higher Education Institutions of China (No. 18KJA110002) and 2018 QingLan Project.

where the initial condition u_0 satisfying $\sup_{x \in \mathbb{R}} \mathbb{E}(|u_0(x)|^p) < \infty$ with $p \geq 2$ and $T > 0$. The notations ν and λ are two positive constants. In STF-SHE (1), ∂_t^β ($0 < \beta < 1$) denotes the Caputo fractional derivative which will be explained later, while $(-\Delta)^{\frac{\alpha}{2}}$ is the generator of the symmetric stable process with $0 < \alpha \leq 2$. The coefficient $\sigma(\cdot) : \mathbb{R} \rightarrow \mathbb{R}$ is a non-random measurable function and $\dot{F}_\rho(t, x)$ denotes the spatially inhomogeneous white noise with intensity measure $dt\rho(dx)$, where ρ denotes a σ -finite positive Radon measure on $\mathbb{R}$. $F_\rho(\cdot, \cdot)$ is the related inhomogeneous Brownian sheet which has been studied in [26] and [27]. One can see more details about the noise $\dot{F}_\rho(t, x)$ and the stochastic process $F_\rho(\cdot, \cdot)$ in Section 2.

The notation $I_t^{1-\beta}$ appearing in STF-SHE (1) is the fractional integral operator defined by

$$I_t^{1-\beta} f(t) := \frac{1}{\Gamma(1-\beta)} \int_0^t \frac{f(r)}{(t-r)^\beta} dr, \quad \beta \in (0, 1), \quad (2)$$

and the fractional derivative operator $\partial_t^\beta = \frac{\partial^\beta}{\partial t^\beta}$ with $\beta \in (0, 1)$ is understood in the *Caputo* sense (see, for example, Caputo [3])

$$\partial_t^\beta u(t, x) = \frac{1}{\Gamma(1-\beta)} \int_0^t \frac{\partial}{\partial r} u(r, x) \frac{dr}{(t-r)^\beta}.$$

As already explained in [18] and [19], due to the appearance of the Caputo fractional derivative ∂_t^β ($0 < \beta < 1$) in STF-SHE (1), we can make use of the *time fractional Duhamel's principle* (one can see [22] and references therein for more details) to get the correct version of STF-SHE (1). If the fractional integral operator $I_t^{1-\beta}$ defined by (2) is not included in STF-SHE (1), by using the fractional Duhamel's principle, the solution to STF-SHE (1) can be given by

$$u(t, x) = \int_{\mathbb{R}} G_t(x-y) u_0(y) dy + \int_0^t \int_{\mathbb{R}} G_{t-s}(x-y) \partial_s^{1-\beta} [\lambda \sigma(u(s, y)) \dot{F}_\rho(s, y)] dy ds.$$

It is not clear what the fractional derivative $\partial_s^{1-\beta} [\sigma(u(s, y)) \dot{F}_\rho(s, y)]$ means. So we can use the fractional integral operator $I_t^{1-\beta}$ to remove the fractional derivative of the noise term $\partial_s^{1-\beta} [\lambda \sigma(u(s, y)) \dot{F}_\rho(s, y)]$. One mainly can use the following property concerning the fractional derivative operator ∂_t^β and fractional integral operator $I_t^{1-\beta}$: for every $\beta \in (0, 1)$, and $g \in L^\infty(\mathbb{R}_+)$ or $g \in C(\mathbb{R}_+)$, $\partial_t^\beta I_t^{1-\beta} g(t) = g(t)$. For more details, one can see [8], [10], [18] and [19] and references therein.

On one hand, the solution to STF-SHE (1) possesses a physical explanation of time fractional stochastic partial differential equations (SPDEs for short) which has been studied in [4]. The time fractional SPDEs studied in this work may arise naturally by considering the heat equation in material with thermal memory. Much effort has been made in order to introduce a rigorous mathematical approach. See, for example Meerschaert *et al* [17] for some aspects of these results. They are also proved to be very useful for modelling in biological physics and other fields (see, for example, [4], [17], [18], [19]).

It might come natural to add a noise to the time fractional heat type equation (3) and study related theoretical and applied problems. In [18], the authors proved the existence and uniqueness

of the mild solution of the corresponding STF-SHE (1) driven by the space-time white noise which is homogeneous in time and space variables. While in [19], for STF-SHE (1) with space-time homogeneous white noise, the authors proved that the absolute moments of the solutions of this equation grow exponentially and the distances to the origin of the farthest high peaks of those moments grow exactly linearly with time. Furthermore, for the STF-SHE (1) with some spatially homogeneous Gaussian noise, the authors in [10] studied the asymptotic behaviour of the corresponding solution with respect to the time variable and the parameter λ . The authors in [8] studied non-linear noise excitation of the STF-SHE (1) in bounded domain driven by two kinds of multiplicative Gaussian noise which is also homogeneous in time and space variables. Furthermore, in [25], the authors used the weak convergence approach to study the large deviation principle for a STF-SHE (1) with additive fractional noise.

We would like to mention some special case of the parameters α, β in STF-SHE (1). If $\alpha = 2, \beta = 1$, the STF-SHE (1) reduces to the standard stochastic heat equation driven by the spatially inhomogeneous white noise. The author in [26], [27] proved that the solution to

$$\frac{\partial}{\partial t} X_t(x) = \frac{1}{2} \Delta X_t(x) + \sqrt{X_t(x)} \dot{F}_\rho(t, x),$$

is the joint continuity density field of a continuous catalytic super-Brownian motion with catalyst $\rho = (\rho(t, dx) : t \geq 0)$. This above equation can be used to describe the evolution of an infinitesimal system (high-density or short-lifetime limit) concerning critical binary branching Brownian particles. This noise $\dot{F}_\rho(t, x)$ is inhomogeneous in space variable and can be connected to the catalytic super-Brownian motion. One can see the references [5], [15], [16], [20], [24], [26], [27] for more details.

During the studies of SPDEs, many interesting analytic properties, such as sample properties, intermittency and noise excitation have been extensively studied (for example, [6], [7], [8], [9], [10], [12], [13], [24] and etc). In [8], [11], [19], [24], the authors studied the intermittency and moment bounds for several classes of SPDE driven by spatially homogeneous Gaussian noise. In particular, in [24], the author proved that the solution of the one-dimensional stochastic heat equation driven by an inhomogeneous Brownian sheet is *weakly full intermittent* (see Section 3 for the definition) based on the estimates of moments of the solution. Inspired by the above results, in this paper, we will extend the results in [24], [26] and [27] to STF-SHE (1). The first novelty in this work is that the noise structure is different from those SPDEs studied in [6], [7], [8], [9], [10], [11], [19] and so on, in which the noise is homogeneous in space variable. Specially, the catalytic measure $\rho(t, dx)$ relative to the inhomogeneous noise $\dot{F}_\rho(t, x)$ is not required to be absolutely continuous with respect to the Lebesgue measure. Secondly, the Green function $G_t(x)$ associated with STF-SHE (1) is much more complex than that of (fractional) stochastic heat equation (cf. [24], [26] and [27]). Moreover the $G_t(x)$ is lack of semigroup property if $\beta \in (0, 1)$. In particular, we need some more technical tricks to prove the Hölder regularity of the solution to STF-SHE (1) with respect to the time and space variables.

The rest of this paper is organized as follows. In Section 2 we recall some preliminaries on the inhomogeneous space-time white noise $\dot{F}_\rho(t, x)$ and the Green function $G_t(x)$ which is the solution

to the fractional type heat equation (3). In Section 3.1, we prove the existence of the mild solution to STF-SHE (1) and also derive an upper bound for the p -th ($p \geq 2$) moment for the solution. Later on in Section 3.2, we prove the lower bound for the second moment for the solution. Thus we can prove that the solution to STF-SHE (1) is weakly full intermittent. Finally some useful estimates are given in the appendix.

2 Preliminaries

In this section let us firstly recall the spatially inhomogeneous white noise $\dot{F}_\rho(t, x)$ appearing in STF-SHE (1) and then list some assumptions on the catalyst $\rho(dx)$, which induces the joint continuity of the solution (see, for example, [16], [24], [26], [27]). Secondly we recall some results about Green function concerning with the fractional type heat equation (3).

2.1 The spatially inhomogeneous white noise

In this subsection, we mainly recall the stochastic integral with respect to spatially inhomogeneous white noise $\dot{F}_\rho(t, x)$. In order to avoid the confusion, we will write $\rho(dt, dx) = dt\rho(dx)$ for simplicity in the next definition, where $\rho(\cdot)$ is a σ -finite positive Randon measure on $\mathbb{R}$. Let $(\Omega, \mathcal{F}, \mathcal{F}_t, P)$ be a complete probability space in this paper, in which the filtration $(\mathcal{F}_t)$ satisfies the usual hypothesis. We firstly recall the definition of the spatially inhomogeneous white noise $\dot{F}_\rho(t, x) := \frac{\partial^2 F_\rho}{\partial t \partial x}(t, x)$ for completeness (see also, for example, [24], [26], [27]).

Definition 1. 1. Let W_ρ be a real-valued random set function on the sets $\mathcal{B}([0, T] \times \mathbb{R})$. Then W_ρ is said to be a Gaussian white noise on $[0, T] \times \mathbb{R} \times \Omega$ with the intensity measure $\rho(dt, dx) := dt\rho(dx)$ if the following two conditions are satisfied

- (a) For any $A \in \mathcal{B}([0, T] \times \mathbb{R})$ with $\rho(A) < +\infty$, the random variable $W_\rho(A)$ is distributed as the centred Gaussian random variable with the variance $\rho(A)$.
- (b) For any disjoint sets $A, B \in \mathcal{B}([0, T] \times \mathbb{R})$ with $\rho(A) < +\infty, \rho(B) < +\infty$, the random variables $W_\rho(A)$ and $W_\rho(B)$ are independent and

$$W_\rho(A \cup B) = W_\rho(A) + W_\rho(B), \quad \text{a.s.}$$

- 2. The two-parameter stochastic process $\{F_\rho(t, x); t \in [0, T], x \in \mathbb{R}\}$ determined by the following relation is said to be the inhomogeneous Brownian sheet.

For any $0 \leq t < t' \leq T, x < x'$, the following holds,

$$W_\rho((t, t'] \times (x, x']) = F_\rho(t', x') - F_\rho(t', x) - F_\rho(t, x') + F_\rho(t, x), \quad \text{a.s.}$$

From the above definition, one calls that $F_\rho(t, x)$ the homogeneous Brownian sheet if the intensity measure $\rho(dt, dx) = dt dx$. While for the general ρ , one can call $F_\rho(t, x)$ the inhomogeneous

Brownian sheet. The noise $\dot{F}_\rho$ appearing in STF-SHE (1) can be regarded as the density of $\{F_\rho(t, x)\}$ in the distribution sense, which is called the spatially inhomogeneous white noise throughout this paper.

According to the theory of stochastic integral introduced in [23], for any predictable stochastic process $u(t, x, \omega)$ defined on $[0, T] \times \mathbb{R} \times \Omega$ satisfying

$$\mathbb{E} \left[\int_0^t \int_{\mathbb{R}} u^2(s, x) ds \rho(dx) \right] < +\infty, \quad t \in [0, T],$$

one can define the stochastic integral of $u \cdot F_\rho = \{(u \cdot F_\rho)(t), t \in [0, T]\}$ by using the well-known Itô's isometry

$$\mathbb{E} [| (u \cdot F_\rho)(t) |^2] = \mathbb{E} \left[\int_0^t \int_{\mathbb{R}} u^2(s, x) ds \rho(dx) \right].$$

As usual, we denote the stochastic integral $(u \cdot F_\rho)(t)$ by (see, for example, [20], [24], [26], [27])

$$(u \cdot F_\rho)(t) := \int_0^t \int_{\mathbb{R}} u(s, x) F_\rho(ds, dx).$$

Moreover, the process $(u \cdot F_\rho)(t)$ is an $\mathcal{F}_t$ -martingale and its quadratic variation process is

$$\langle (u \cdot F_\rho)(\cdot) \rangle_t = \int_0^t \int_{\mathbb{R}} u^2(s, x) ds \rho(dx), \quad t \geq 0.$$

Inspired by the assumptions in [20], [24] and [27], in order to study the moment bounds and intermittent properties of the solution of STF-SHE (1), some mild assumptions on the catalyst ρ are required. For our aim, throughout this paper, we impose the following assumption **(A)** on the catalyst ρ (also appearing in the studies of [20], [24], [26] and [27]).

(A.1) The catalyst ρ is a σ -finite positive Randon measure on $\mathbb{R}$ such that

$$\sup_{x \in \mathbb{R}} \rho(B(x, 1)) < +\infty.$$

Here, for each $r > 0$, $B(x, r)$ denotes the r -neighborhood of $x \in \mathbb{R}$, that is, $B(x, r) = \{y \in \mathbb{R} : y \in (x - r, x + r)\}$.

(A.2) There exist a positive constant $C_u > 0$ and some $\bar{\delta} \in (0, 2)$ such that for any $r \in (0, 1]$,

$$\sup_{x \in \mathbb{R}} \rho(B(x, r)) \leq C_u r^{\bar{\delta}}.$$

Remark 1. 1. Under assumptions **(A.1)** and **(A.2)**, the author in [27] proved that the catalytic super-Brownian motion induced a jointly continuous density field. Moreover the density field can be characterized as the unique solution to STF-SHE (1) with $\beta = 1$ and $\alpha = 2$ and $\sigma(u) = \sqrt{u}$.

2. One can consult the Example 1.1 and Example 1.2 in [27] in which the catalyst ρ satisfy the assumptions **(A.1)** and **(A.2)**. Here we omit the details about the two examples. However we should point out that in Example 1.2 in [27], the measure ρ is not absolutely continuous with respect to the Lebesgue measure.

In this paper, we will also impose the following assumption **(B)** on the diffusive coefficient $\sigma(\cdot)$.

(B) The coefficient σ is Lipschitz continuous on $\mathbb{R}$, that is, there exists a constant $K_u > 0$ such that for all $u_1, u_2 \in \mathbb{R}$,

$$|\sigma(u_1) - \sigma(u_2)| \leq K_u |u_1 - u_2|.$$

For each $\theta > 0$ and $p \geq 2$, let us denote by $\mathbb{B}_{p,\theta}$ the class of all $\mathcal{F}_t$ -adapted and continuous random field $u = \{u(t, x); t \in [0, T], x \in \mathbb{R}\}$ satisfying

$$\sup_{t \in [0, T]} \sup_{x \in \mathbb{R}} \mathbb{E}[e^{-\theta t} |u(t, x)|^p] < +\infty.$$

Let us consider the norm $\|\cdot\|_{p,\theta}$ on $\mathbb{B}_{p,\theta}$ defined by

$$\|u\|_{p,\theta} = \left(\sup_{t \in [0, T]} \sup_{x \in \mathbb{R}} \mathbb{E}[e^{-\theta t} |u(t, x)|^p] \right)^{1/p}, \quad u \in \mathbb{B}_{p,\theta}.$$

Then it is easy to see that $(\mathbb{B}_{p,\theta}, \|\cdot\|_{p,\theta})$ is a Banach space (cf. [24]).

Next we also want to study the lower moment bounds for the solution to STF-SHE (1), we also need the following assumption **(C)** on the diffusion coefficient σ and the catalyst ρ .

(C.1) Assume that $\sigma(0) = 0$ and there exists a constant $K_l > 0$ such that

$$K_l |u| \leq |\sigma(u)|, \quad u \in \mathbb{R}.$$

It is natural to require that $K_l \leq K_u$, where K_u is the Lipschitz constant appearing in the assumption **(B)**.

(C.2) There exists $\underline{\delta} \in (0, 2)$ satisfying $\bar{\delta} < \underline{\delta}$ and a constant $C_l > 0$ such that

$$\inf_{x \in \mathbb{R}} \rho(B(x, r)) \geq C_l r^{\underline{\delta}}, \quad r \in (0, 1].$$

In particular, we have that $\inf_{x \in \mathbb{R}} \rho(B(x, 1)) > 0$.

2.2 Some properties of Green function

Assuming $\nu > 0$, we limit our consideration to the following parameter ranges of β and α (i.e., $\beta \in (0, 1)$ and $\alpha \in (0, 2]$) since we plan to use some properties of fundamental solution (or Green function) $G_t(x)$ of the fractional heat type equation

$$\partial_t^\beta G_t(x) = -\nu(-\Delta)^{\frac{\alpha}{2}} G_t(x). \quad (3)$$

In fact, the deterministic counterparts of STF-SHE (1) (that is Eq. (3)) have received many attentions and are related with anomalous diffusions or diffusions in non-homogeneous media with random fractal structures (cf. [2], [17] and etc). It is known that $G_t(x)$ is the density function of $X(E_t)$, where X is a symmetric α -stable Lévy process in $\mathbb{R}$ and E_t is the first passage time of an β -stable subordinator $D = \{D_r, r \geq 0\}$, or the inverse stable subordinator of index β . According to [10], [18], [19], we know that

$$G_t(x) = \int_0^\infty p_{X(s)}(x) h_{E_t}(s) ds, \quad (4)$$

where $p_{X(s)}(x)$ and $h_{E_t}(s)$ are the densities of $X(s)$ and E_t , respectively. Moreover the Fourier transform of $p_{X(s)}(x)$ is given by

$$\widehat{p_{X(s)}}(\xi) = e^{-s\nu|\xi|^\alpha},$$

and

$$h_{E_t}(x) = t\beta^{-1}x^{-1-\frac{1}{\beta}}g_\beta(tx^{-1/\beta}),$$

where $g_\beta(\cdot)$ is the density function of D_1 . The function $g_\beta(u)$ (cf. Mijena and Nane [18]) is infinitely differentiable on the entire real line, with $g_\beta(u) = 0$ for $u \leq 0$.

3 Moments bounds and intermittency

In this section we will give the first result which is concerned with the existence and uniqueness of the solution $u(t, x)$ to STF-SHE (1) in some Banach space $\mathbb{B}_{p,\theta}$ and prove that the upper bound of p -th ($p \geq 2$) moments of the solution $u(t, x)$ grows at most exponentially. We also prove that the solution is Hölder continuous with respect to the time and space variables. For the second result, we will prove the uniformly lower Lyapunov's exponent of the solution $u(t, x)$ to STF-SHE (1), which implies that the solution $u(t, x)$ grows at least exponentially. Then we can prove that the solution to STF-SHE (1) is *weakly full intermittent*.

For our purpose, let us recall the mild solution in the sense of the random field (i.e. [23]).

Definition 2. An $\mathcal{F}_t$ -adapted random field $u = \{u(t, x) : \equiv u(t, x, \omega); t \in [0, T], x \in \mathbb{R}, \omega \in \Omega\}$ is said to be a mild solution of STF-SHE (1) with the initial condition u_0 if

$$\sup_{t \in [0, T]} \sup_{x \in \mathbb{R}} \mathbb{E}[|u(t, x)|^p] < \infty, \quad p \geq 2,$$

and the following integral equation is satisfied with probability one

$$u(t, x) = \int_{\mathbb{R}} G_t(x - y) u_0(y) dy + \lambda \int_0^t \int_{\mathbb{R}} \sigma(u(s, y)) G_{t-s}(x - y) F_\rho(ds, dy), \quad (5)$$

where $G_t(x)$ denotes the fundamental solution of Eq. (3).

3.1 Upper bounds for the p -th moment

Let us formulate the first main result in this paper, which is about the existence and uniqueness of the solution in some Banach space $\mathbb{B}_{p,\theta}$. Next we show that the p -th ($p \geq 2$) moment of the solution to the STF-SHE (1) grows at most exponentially. We also study the Hölder continuity of the solution. We firstly consider $\nu = 1$ and $\lambda = 1$ for simplicity. The main results in this subsection are formulated as the following theorem.

Theorem 1. *Let the assumptions (A.1), (A.2) and (B) be satisfied and assume that $1 < \alpha \leq 2$ and $0 < \beta < \min\{1, \frac{\alpha}{2-\delta}\}$. If the initial condition $u_0(x)$ satisfying $\sup_{x \in \mathbb{R}} \mathbb{E}(|u_0(x)|^p) < \infty$ with $p \geq 2$, then the following hold:*

1. *For each $p \geq 2$, there exists some $\theta_0 > 0$ depending on $p, \bar{\delta}$ such that for any $\theta > \theta_0$, the STF-SHE (1) has a unique mild solution $u(t, x)$ in the Banach space $\mathbb{B}_{p,\theta}$.*
2. *There are two positive constants $C_1 > 0$ and $C_2 > 0$ independent of K_u such that*

$$\sup_{x \in \mathbb{R}} \mathbb{E}[|u(t, x)|^p] \leq C_1^{\frac{p}{2}} \left(1 + \sup_{x \in \mathbb{R}} \mathbb{E}[|u_0(x)|^p] \right) \exp \left\{ C_2 K_u^{\frac{2\alpha}{\alpha+\beta(\bar{\delta}-2)}} p^{1+\frac{\alpha}{\alpha+\beta(\bar{\delta}-2)}} t \right\}, \quad (6)$$

simultaneously for $p \geq 2$ and $t > 0$.

3. *For any fixed $T > 0$, the solution $u = \{u(t, x), (t, x) \in (0, T] \times \mathbb{R}\}$ to STF-SHE (1) is μ -Hölder continuous in temporal variable t and respectively, κ -Hölder continuous in spatial variable $x \in \mathbb{R}$ for any $\mu \in \left(0, \frac{\alpha+\beta(\bar{\delta}-2)}{2\alpha}\right)$ and $\kappa \in \left(0, \min\left\{\frac{\alpha+\beta(\bar{\delta}-2)}{2\beta}, 1\right\}\right)$*

Remark 2. 1. *The assumption that $0 < \beta < \min\{1, \frac{\alpha}{2-\delta}\}$ is important in the above theorem. The order of fractional operator with respect to the space variable is bigger than that of the time variable. This coincides with the case that if $\beta = 1$, the STF-SHE (1) has a unique mild solution if $1 < \alpha \leq 2$ which also implies that $\beta < \alpha$.*

2. *If the catalyst $\rho(dx) = cdx$ with c a positive constant and $\beta \in (0, 1)$, one obtains that the exponents $\frac{2\alpha}{\alpha+\beta(\bar{\delta}-2)}$ and $1 + \frac{\alpha}{\alpha+\beta(\bar{\delta}-2)}$ in (6) reduces to the quantities $\frac{2\alpha}{\alpha-\beta}$ and $\frac{2\alpha-\beta}{\alpha-\beta}$, which coincided the results obtained in [9] (Theorem 1.3), [10] (Theorem 1.4) or [18] (Theorem 2) for the fractional stochastic heat equation. Furthermore, in this case, the orders μ, κ of Hölder continuity of the solution $u(t, x)$ reduce to $\frac{\alpha-\beta}{2\alpha}$ (which has been proved by Theorem 2 in [18]) and $\frac{\alpha-\beta}{2\beta}$. Especially, the result about the order of spatial Hölder continuity is new.*
3. *If the catalyst $\rho(dx)$ satisfies the assumptions (A.1), (A.2), $\alpha = 2$ and β is close to 1, one obtains that the exponents $\frac{2\alpha}{\alpha+\beta(\bar{\delta}-2)}$ and $1 + \frac{\alpha}{\alpha+\beta(\bar{\delta}-2)}$ in (6) reduces to the quantities $\frac{4}{\bar{\delta}}$ and $1 + \frac{2}{\bar{\delta}}$, which coincides the results (Theorem 2.3) obtained in [24] for the stochastic heat equation with strongly inhomogeneous space-time white noise. In this case, the orders μ and κ of Hölder continuity of the solution $u(t, x)$ is less than $\frac{\bar{\delta}}{4}$ and $\frac{\bar{\delta}}{2}$ which coincide with point*

(1) in Theorem 2.3 in [24]. Furthermore when $\bar{\delta} = 1$, the order of Hölder regularity turn to be less $\frac{1}{4}$ and $\frac{1}{2}$, respectively, which had been proved in studying stochastic heat equation with Gaussian space-time white noise by [23].

We begin with some lemmas and then formulate the proof of Theorem 1. For $u(t, x)$ in $\mathbb{B}_{p,\theta}$, let us denote by $\mathbb{S}u(t, x)$ the stochastic convolution

$$\mathbb{S}u(t, x) = \int_0^t \int_{\mathbb{R}} G_{t-s}(x - y) u(s, y) F_{\rho}(ds, dy), \quad t \in [0, T], x \in \mathbb{R},$$

in this section. Let us firstly recall the following stochastic Young type inequality which has been proved in [24].

Lemma 1. *For any $u \in \mathbb{B}_{p,\theta}$ and $p \geq 2$, we have that*

$$\mathbb{E} [|\mathbb{S}u(t, x)|^p]^{\frac{2}{p}} \leq 4p \int_0^t \int_{\mathbb{R}} G_{t-s}^2(x - y) \mathbb{E}[|u(s, y)|^p]^{\frac{2}{p}} ds \rho(dy). \quad (7)$$

Then we have the following result concerning the stochastic convolution $\mathbb{S}u(t, x)$ of solution $u(t, x)$ to STF-SHE (1).

Lemma 2. *For any $u \in \mathbb{B}_{p,\theta}$, we have that $\mathbb{S}u \in \mathbb{B}_{p,\theta}$. More precisely, there exists a constant $C_3 > 0$ appearing in (33) which is independent of $p, \alpha, \beta, \theta, \bar{\delta}$ such that for any $u \in \mathbb{B}_{p,\theta}$*

$$\|\mathbb{S}u\|_{p,\theta}^2 \leq 4p C_3 C_{\bar{\delta}}^*(\alpha, \beta, \theta) \|u\|_{p,\theta}^2, \quad (8)$$

where $C_{\bar{\delta}}^*(\alpha, \beta, \theta)$ is defined by (34).

Proof: By the definition of $\|\cdot\|_{p,\theta}$ and Lemma 1, one obtains that

$$\begin{aligned} \|\mathbb{S}u\|_{p,\theta}^2 &= \sup_{t \in [0, T]} \sup_{x \in \mathbb{R}} \exp \left\{ -\frac{2\theta t}{p} \right\} \mathbb{E}[|\mathbb{S}u(t, x)|^p]^{\frac{2}{p}} \\ &\leq 4p \sup_{t \in [0, T]} \sup_{x \in \mathbb{R}} \exp \left\{ -\frac{2\theta t}{p} \right\} \int_0^t \int_{\mathbb{R}} G_{t-s}^2(x - y) \mathbb{E}[|u(s, y)|^p]^{\frac{2}{p}} \rho(dy) ds \\ &\leq 4p \|u\|_{p,\theta}^2 \sup_{t \in [0, T]} \sup_{x \in \mathbb{R}} \exp \left\{ -\frac{2\theta t}{p} \right\} \int_0^t \int_{\mathbb{R}} G_{t-s}^2(x - y) \exp \left\{ \frac{2\theta s}{p} \right\} \rho(dy) ds \\ &\leq 4p \|u\|_{p,\theta}^2 \sup_{t \in [0, T]} \sup_{x \in \mathbb{R}} \int_0^t \int_{\mathbb{R}} G_s^2(x - y) \exp \left\{ -\frac{2\theta s}{p} \right\} \rho(dy) ds. \end{aligned}$$

Consequently, we can easily obtain the desired result by using Lemma 9 proved in the appendix. ■

We also have the following

Lemma 3. For any $p \geq 2$ and $\theta > 0$, there exists a constant $C'_3 > 0$ which is independent of $\alpha, \beta, \bar{\delta}, \theta, p$ such that for any $u, v \in \mathbb{B}_{p,\theta}$

$$\|\mathbb{S}u - \mathbb{S}v\|_{p,\theta}^2 \leq 4pC'_3C_{\bar{\delta}}^*(\alpha, \beta, \theta)\|u - v\|_{p,\theta}^2, \quad (9)$$

holds, where $C_{\bar{\delta}}^*(\alpha, \beta, \theta)$ is defined (34).

Proof: The proof is essentially the same as that of Lemma 2. In fact, by the definition of the stochastic convolution $\mathbb{S}u(t, x)$, we see that

$$\begin{aligned} \|\mathbb{S}u - \mathbb{S}v\|_{p,\theta}^2 &= \sup_{t \in [0, T]} \sup_{x \in \mathbb{R}} \exp \left\{ -\frac{2\theta t}{p} \right\} \mathbb{E}[|\mathbb{S}u(t, x) - \mathbb{S}v(t, x)|^p]^{\frac{2}{p}} \\ &= \sup_{t \in [0, T]} \sup_{x \in \mathbb{R}} \exp \left\{ -\frac{2\theta t}{p} \right\} \mathbb{E} \left[\left| \int_0^t \int_{\mathbb{R}} G_{t-s}(x - y)(u(s, y) - v(s, y)) F_{\rho}(ds, dy) \right|^p \right]^{\frac{2}{p}} \end{aligned}$$

Consequently, we can easily obtain the desired result by using Lemma 2, since $\mathbb{B}_{p,\theta}$ is a Banach space. ■

Now let us assume $0 < \beta < \frac{\alpha}{2-\bar{\delta}}$, we would like to show that, for each $u \in \mathbb{B}_{p,\theta}$, the stochastic convolution $\mathbb{S}u(t, x)$ has the following moment estimate of the increments with respect to the time and space variables. This result can be stated as follows.

Lemma 4. Assume that $0 < \beta < \min\{1, \frac{\alpha}{2-\bar{\delta}}\}$ and let $T > 0$ be fixed. If for any $p \geq 2$,

$$\sup_{t \in (0, T]} \sup_{x \in \mathbb{R}} \mathbb{E}[|u(t, x)|^p] < \infty,$$

then for any $0 < t_1, t_2 \leq T$ and $x_1, x_2 \in \mathbb{R}$, we have that for some constant $C > 0$

$$\begin{aligned} &\mathbb{E}[|\mathbb{S}u(t_2, x_2) - \mathbb{S}u(t_1, x_1)|^p] \\ &\leq C2^{p-1}(4p)^{p/2} \sup_{t \in (0, T]} \sup_{x \in \mathbb{R}} \mathbb{E}[|u(t, x)|^p] (|t_2 - t_1|^{2\mu} + |x_2 - x_1|^{2\kappa})^{\frac{p}{2}}, \end{aligned}$$

with $0 < \mu < \frac{\alpha + \beta(\bar{\delta} - 2)}{2\alpha}$ and $0 < \kappa < \min\{\frac{\alpha + \beta(\bar{\delta} - 2)}{2\beta}, 1\}$. The Kolmogorov continuity theorem yields that the stochastic convolution $\mathbb{S}u(t, x)$ is μ -Hölder continuity in temporal variable, and κ -Hölder continuous in spatial variable.

Proof: Assume that $p \geq 2$ and $0 < t_1 < t_2 \leq T$ for any $T > 0$. By using the Burkholder inequality,

we have that, for every $x_1, x_2 \in \mathbb{R}$,

$$\begin{aligned}
& \mathbb{E} [|Su(t_2, x_2) - Su(t_1, x_1)|^p] \\
& \leq 2^{p-1} \mathbb{E} \left[\left| \int_0^{t_1} \int_{\mathbb{R}} (G_{t_2-s}(x_2 - y) - G_{t_1-s}(x_1 - y)) u(s, y) F_{\rho}(ds, dy) \right|^p \right] \\
& \quad + 2^{p-1} \mathbb{E} \left[\left| \int_{t_1}^{t_2} \int_{\mathbb{R}} G_{t_2-s}(x_2 - y) u(s, y) F_{\rho}(ds, dy) \right|^p \right] \\
& \leq 2^{p-1} (4p)^{p/2} \sup_{s \in (0, T]} \sup_{y \in \mathbb{R}} \mathbb{E} [|u(s, y)|^p] \left(\int_0^{t_1} \int_{\mathbb{R}} (G_{t_2-s}(x_2 - y) - G_{t_1-s}(x_1 - y))^2 \rho(dy) ds \right)^{p/2} \\
& \quad + 2^{p-1} (4p)^{p/2} \sup_{s \in (0, T]} \sup_{y \in \mathbb{R}} \mathbb{E} [|u(s, y)|^p] \left(\int_{t_1}^{t_2} \int_{\mathbb{R}} G_{t_2-s}^2(x_2 - y) \rho(dy) ds \right)^{p/2}
\end{aligned}$$

Then by using the proofs for the upper bounds of $\mathcal{L}, \mathcal{M}, \mathcal{N}$ defined in Lemma 11, one can conclude the proof of this lemma. ■

Now let us give the proof of Theorem 1 based on the above preparations.

Proof: [Proof of Theorem 1] Let us first state the proof of the first part (I) in Theorem 1. By the assumptions on the initial condition $u_0(x)$, we can easily get the first term in the right hand of (5)

$$\mathbb{T}_{0,t} u_0(x) := G_t * u_0(x) = \int_{\mathbb{R}} G_t(x - y) u_0(y) dy,$$

is an element of $\mathbb{B}_{p,\theta}$ for any $\theta > 0$. In fact by the contractivity of the mapping $\mathbb{T}_{0,t} u_0(x)$, we have that

$$\begin{aligned}
\|\mathbb{T}_{0,t} u_0(x)\|_{p,\theta}^p &= \sup_{(t,x) \in (0,T] \times \mathbb{R}} \mathbb{E} \left[e^{-\theta t} \left| \int_{\mathbb{R}} G_t(x - y) u_0(y) dy \right|^p \right] \\
&\leq \sup_{(t,x) \in (0,T] \times \mathbb{R}} \mathbb{E} \left[e^{-\theta t} \int_{\mathbb{R}} G_t(x - y) |u_0(y)|^p dy \right] \\
&\leq \sup_{y \in \mathbb{R}} \mathbb{E} [|u_0(y)|^p] \sup_{t \in (0,T]} e^{-\theta t} \\
&= \sup_{y \in \mathbb{R}} \mathbb{E} [|u_0(y)|^p].
\end{aligned} \tag{10}$$

Now let us consider the following mapping $\mathbb{T}$ defined by

$$\mathbb{T}u(t, x) = \mathbb{T}_{0,t} u_0(x) + \mathbb{S}\sigma(u(t, x)), \quad u \in \mathbb{B}_{p,\theta}. \tag{11}$$

Then by Lemma 2 and inequality (10), it is easy to know that the operator $\mathbb{T}$ defined by (11) from $\mathbb{B}_{p,\theta}$ to $\mathbb{B}_{p,\theta}$ is surjective. In fact, using Lemma 2 and the Minkowski inequality twice, we can obtain

that for $u \in \mathbb{B}_{p,\theta}$

$$\begin{aligned} \|\mathbb{T}u(t, x)\|_{p,\theta}^2 &\leq 2\|\mathbb{T}_{0,t}u_0(x)\|_{p,\theta}^2 + 2\|\mathbb{S}\sigma(u(t, x))\|_{p,\theta}^2 \\ &\leq 2\sup_{y \in \mathbb{R}} \mathbb{E}[|u_0(y)|^p]^{\frac{2}{p}} + 8pC_3C_{\bar{\delta}}^*(\alpha, \beta, \theta) (|\sigma(0)|^2 + K_u^2\|u\|_{p,\theta}^2), \end{aligned} \quad (12)$$

where C_3 and $C_{\bar{\delta}}^*(\alpha, \beta, \theta)$ are the constants appearing in Lemma 2. Thus one gets that $\mathbb{T}u \in \mathbb{B}_{p,\theta}$.

Next let us show the contractivity of the mapping $\mathbb{T}$ on $\mathbb{B}_{p,\theta}$ when θ is large enough. In fact, by using the Lipschitz continuity of σ , see also the assumption **(B)**, and by using Lemma 3, we have that for any $u, v \in \mathbb{B}_{p,\theta}$,

$$\begin{aligned} \|\mathbb{T}u - \mathbb{T}v\|_{p,\theta}^2 &\leq \|\mathbb{S}\sigma(u(t, x)) - \mathbb{S}\sigma(v(t, x))\|_{p,\theta}^2 \\ &\leq 4pC_3' C_{\bar{\delta}}^*(\alpha, \beta, \theta) K_u^2 \|u - v\|_{p,\theta}^2. \end{aligned} \quad (13)$$

Let us write

$$\theta_0 = \inf \{ \theta > 0 : 4pC_3' C_{\bar{\delta}}^*(\alpha, \beta, \theta) K_u^2 < 1 \}.$$

Here by convention, we set $\inf \{\emptyset\} = \infty$. Since $\bar{\delta} \in (0, 1)$ and $0 < \beta < \min\{1, \frac{\alpha}{2-\bar{\delta}}\}$, we know that $C_{\bar{\delta}}^*(\alpha, \beta, \theta)$ given by (34) is decreasing in θ and $\lim_{\theta \rightarrow \infty} C_{\bar{\delta}}^*(\alpha, \beta, \theta) = 0$. Therefore, we have that θ_0 exists and for all $\theta > \theta_0$, one has $4pC_3' C_{\bar{\delta}}^*(\alpha, \beta, \theta) K_u^2 < 1$. Hence, from (13), it is known that the mapping $\mathbb{T} : \mathbb{B}_{p,\theta} \rightarrow \mathbb{B}_{p,\theta}$ is contractive if $\theta > \theta_0$. Consequently, by the fixed point theorem, the STF-SHE (1) has a unique mild solution in $\mathbb{B}_{p,\theta}$ with $\theta > \theta_0$.

From now on, we will formulate the proof of the second point. Firstly, recalling that $\bar{\delta} \in (0, 1]$, for each $p \geq 2$, we can choose a positive constant $C > 0$ large enough depending only on the constants C_3 and C_3' appearing in (12) and (13) such that whenever

$$\theta = CK_u^{\frac{2\alpha}{\alpha+\beta(\bar{\delta}-2)}} p^{1+\frac{\alpha}{\alpha+\beta(\bar{\delta}-2)}} > \theta_0,$$

we have $8pC_3C_{\bar{\delta}}^*(\alpha, \beta, \theta)K_u^2 \leq \frac{1}{2}$. Let us also set $C' = 8pC_3C_{\bar{\delta}}^*(\alpha, \beta, \theta)|\sigma(0)|^2$. Note that for the special θ defined above, $pC_{\bar{\delta}}^*(\alpha, \beta, \theta)$ is bounded. On the other hand, the solution $u(t, x)$ to STF-SHE (1) is the unique fixed point of the mapping $\mathbb{T}$ from $\mathbb{B}_{p,\theta}$ into $\mathbb{B}_{p,\theta}$, we know that

$$\mathbb{T}u(t, x) = u(t, x), \quad \text{a.s.}$$

Thus from (12), we see that for the case $\theta = CK_u^{\frac{2\alpha}{\alpha+\beta(\bar{\delta}-2)}} p^{1+\frac{\alpha}{\alpha+\beta(\bar{\delta}-2)}}$

$$\|u\|_{p,\theta}^2 \leq 2\sup_{x \in \mathbb{R}} \mathbb{E}[|u_0(x)|^p]^{\frac{2}{p}} + C' + \frac{1}{2}\|u\|_{p,\theta}^2,$$

and then we have

$$\|u\|_{p,\theta}^2 \leq 4\sup_{x \in \mathbb{R}} \mathbb{E}[|u_0(x)|^p]^{\frac{2}{p}} + 2C'.$$

Consequently, from the definition of the norm $\|\cdot\|_{p,\theta}^2$, we can easily deduce that for any $p \geq 2$, there exists a positive constant $C > 0$ depending only on $\sigma(0)$ and $C_{\bar{\delta}}^*(\alpha, \beta, \theta)$ such that

$$\mathbb{E}[|u(t, x)|^p] \leq \left(4 \sup_{x \in \mathbb{R}} \mathbb{E}[|u_0(x)|^p]^{\frac{2}{p}} + 2C' \right) \exp \left\{ C K_u^{\frac{2\alpha}{\alpha+\beta(\bar{\delta}-2)}} p^{1+\frac{\alpha}{\alpha+\beta(\bar{\delta}-2)}} t \right\},$$

which implies the desired result (6).

For the third point of this theorem, we will prove the Hölder continuity of the solution to STF-SHE (1) by using the Kolmogorov's continuity theorem. First of all, let us consider the case for the temporal variable t . For any $0 < t_1 < t_2 \leq T$ with $T > 0$ and any $x \in \mathbb{R}$, we consider the increment $\mathbb{E}[|u(t_2, x) - u(t_1, x)|^p]$ for $p \geq 2$. We have

$$\begin{aligned} u(t_2, x) - u(t_1, x) &= \int_{\mathbb{R}} [G_{t_2}(x - y) - G_{t_1}(x - y)] u_0(y) dy \\ &\quad + \int_0^{t_1} \int_{\mathbb{R}} [G_{t_2-s}(x - y) - G_{t_1-s}(x - y)] \sigma(u(s, y)) F_{\rho}(ds, dy) \\ &\quad + \int_{t_1}^{t_2} \int_{\mathbb{R}} G_{t_2-s}(x - y) \sigma(u(s, y)) F_{\rho}(ds, dy). \end{aligned}$$

For the first term $\int_{\mathbb{R}} [G_{t_2}(x - y) - G_{t_1}(x - y)] u_0(y) dy$ in the above expression, one can get that by using mean-value theorem, for ξ between t_1 and t_2 ,

$$\begin{aligned} &\mathbb{E} \left| \int_{\mathbb{R}} [G_{t_2}(x - y) - G_{t_1}(x - y)] u_0(y) dy \right|^p \\ &\leq \sup_{y \in \mathbb{R}} \mathbb{E}[|u_0(y)|^p] \left(\int_{\mathbb{R}} |G_{t_2}(x - y) - G_{t_1}(x - y)| dy \right)^p \\ &= \sup_{y \in \mathbb{R}} \mathbb{E}[|u_0(y)|^p] |t_2 - t_1|^p \left| \int_{\mathbb{R}} \frac{\partial}{\partial t} G_t(x - y) |_{t=\xi} dy \right|^p \\ &\leq C |t_2 - t_1|^p, \end{aligned}$$

by using the bound $\frac{\partial}{\partial t} G_t(x - y) \leq \frac{C}{t} G_t(x - y)$ proved in [1] and the integral $\left| \int_{\mathbb{R}} G_t(x - y) dy \right|^p$ is bounded.

We will therefore look at the remaining two terms in the difference $u(t_2, x) - u(t_1, x)$. Recall that we have showed that $\mathbb{E}[|u(t, x)|^p]$ is bounded with $t \in (0, T]$ for any $T > 0$ in the proof of the second point of this theorem. We then use assumption **(B)** together with Lemma 4 to write

$$\mathbb{E} \left[\left| \int_0^{t_1} \int_{\mathbb{R}} [G_{t_2-s}(x - y) - G_{t_1-s}(x - y)] \sigma(u(s, y)) F_{\rho}(ds, dy) \right|^p \right] \leq C (t_2 - t_1)^{p\mu},$$

with $0 < \mu < \frac{\alpha+\beta(\bar{\delta}-2)}{2\alpha}$. Similarly, by following the analogous arguments in the proof of Lemma 9,

we have

$$\begin{aligned}
& \mathbb{E} \left[\left| \int_{t_1}^{t_2} \int_{\mathbb{R}} G_{t_2-s}(x-y) \sigma(u(s,y)) F_{\rho}(ds, dy) \right|^p \right] \\
& \leq C \mathbb{E} \left[\left| \int_{t_1}^{t_2} \int_{\mathbb{R}} G_{t_2-s}^2(x-y) \rho(ds) dy \right|^{p/2} \right] \\
& \leq C \int_{t_1}^{t_2} (t_2-s)^{-\frac{2\beta}{\alpha}} \int_{\mathbb{R}} 1_{\{|x-y| \leq (t_2-s)^{\frac{\beta}{\alpha}}\}} \rho(dy) ds \\
& \quad + C \int_{t_1}^{t_2} (t_2-s)^{2\beta} \int_{\mathbb{R}} \frac{1}{|x-y|^{2(1+\alpha)}} 1_{\{|x-y| > (t_2-s)^{\frac{\beta}{\alpha}}\}} \rho(dy) ds \\
& \leq C(t_2-t_1)^{p\mu},
\end{aligned}$$

with $0 < \mu < \frac{\alpha+\beta(\bar{\delta}-2)}{2\alpha}$. Then combining the above estimates for $p \geq 2$, we obtain that

$$\mathbb{E}[|u(t_2, x) - u(t_1, x)|^p] \leq C(t_2 - t_1)^{p\mu},$$

with $0 < \mu < \frac{\alpha+\beta(\bar{\delta}-2)}{2\alpha}$.

Secondly, let us consider the case for the spatial variable x . For any $0 < t \leq T$ with $T > 0$ and any $x_1, x_2 \in \mathbb{R}$, we consider the increment $\mathbb{E}[|u(t, x_2) - u(t, x_1)|^p]$ for $p \geq 2$. We have

$$\begin{aligned}
u(t, x_2) - u(t, x_1) &= \int_{\mathbb{R}} [G_t(x_2 - y) - G_t(x_1 - y)] u_0(y) dy \\
& \quad + \int_0^t \int_{\mathbb{R}} [G_{t-s}(x_2 - y) - G_{t-s}(x_1 - y)] \sigma(u(s, y)) F_{\rho}(ds, dy)
\end{aligned}$$

For the first term $\int_{\mathbb{R}} [G_t(x_2 - y) - G_t(x_1 - y)] u_0(y) dy$ in the above expression, one can get that

$$\begin{aligned}
& \mathbb{E} \left| \int_{\mathbb{R}} [G_t(x_2 - y) - G_t(x_1 - y)] u_0(y) dy \right|^p \\
& \leq \sup_{y \in \mathbb{R}} \mathbb{E} |u_0(y)|^p \left(\int_{\mathbb{R}} [G_t(x_2 - y) - G_t(x_1 - y)] dy \right)^p \\
& = C |x_2 - x_1|^{p\varrho} t^{-\frac{p\varrho\beta}{\alpha}} \\
& \leq C |x_2 - x_1|^{p\varrho}, \quad \text{with } \varrho < \alpha,
\end{aligned}$$

where we have used the estimate (2.15) in [1].

We will therefore look at higher moments of the last term. Recall that we have proved that $\mathbb{E}[|u(t, x)|^p]$ is bounded with $t \in (0, T]$ for any $T > 0$ in the second point of this theorem. We then use assumption **(B)** together with Lemma 4 to write

$$\mathbb{E} \left[\left| \int_0^t \int_{\mathbb{R}} [G_{t-s}(x_2 - y) - G_{t-s}(x_1 - y)] \sigma(u(s, y)) F_{\rho}(ds, dy) \right|^p \right] \leq C |x_2 - x_1|^{p\kappa},$$

with $0 < \kappa < \frac{\alpha+\beta(\bar{\delta}-2)}{2\beta}$. Then combine the above two estimates, we see that for $p \geq 2$

$$\mathbb{E}[|u(t, x_2) - u(t, x_1)|^p] \leq C|x_2 - x_1|^{p\kappa},$$

with $0 < \kappa < \min\left\{\frac{\alpha+\beta(\bar{\delta}-2)}{2\beta}, 1\right\}$.

For $p \geq 2$ and any fixed $T > 0$, let $0 < t_1 < t_2 \leq T$ and $x_1, x_2 \in \mathbb{R}$, one get that

$$\begin{aligned} \mathbb{E}[|u(t_2, x_2) - u(t_1, x_1)|^p] &\leq C_p \mathbb{E}[|u(t_2, x_2) - u(t_1, x_2)|^p] + C_p \mathbb{E}[|u(t_1, x_2) - u(t_1, x_1)|^p] \\ &\leq C(|t_2 - t_1|^{p\mu} + |x_2 - x_1|^{p\kappa}), \end{aligned}$$

with $0 < \mu < \frac{\alpha+\beta(2-\bar{\delta})}{2\alpha}$ and $0 < \kappa < \min\left\{\frac{\alpha+\beta(2-\bar{\delta})}{2\beta}, 1\right\}$. Thus we can conclude the third point of this theorem by applying Kolmogorov's continuity theorem. ■

3.2 Lower bound for the second moment

In this subsection, we are going to consider the uniformly lower Lyapunov exponent $\underline{\mathcal{L}}(p, x)$, $p \geq 2$ of the solution $u(t, x)$ to STF-SHE (1), which implies that the solution $u(t, x)$ grows at least exponentially. In fact, we can prove that $\inf_{x \in \mathbb{R}} \underline{\mathcal{L}}(2, x) > 0$ and then, in particular, we prove that the solution $u(t, x)$ exhibits the weakly full intermittent property under the assumption (C).

From now on, we will assume that the initial condition u_0 is non-negative. Then we can state the main theorem in this subsection as follows.

Theorem 2. Assume that $0 < \beta < \min\{1, \frac{\alpha}{2-\bar{\delta}}\}$ and the assumptions (A.1), (A.2), (B) and (C) are satisfied. Then the following two statements hold.

1. If the initial condition $u_0(x)$ is uniformly bounded from below by $\zeta > 0$, i.e. $\inf_{x \in \mathbb{R}} u_0(x) \geq \zeta$, then for large enough $T > 1$, there exist two constants $\vartheta_0 > 0, \tilde{\vartheta}_0 > 0$ independent of K_l such that

$$\inf_{x \in \mathbb{R}} \mathbb{E}[|u(t, x)|^2] \geq \zeta^2 \exp\{\vartheta_0 K_l^2 t\}, \quad (t, x) \in (0, T] \times \mathbb{R}, \quad (14)$$

and respectively for any $\delta \in (0, 1)$ and $t \geq T$

$$\inf_{x \in \mathbb{R}} \mathbb{E}[|u(t, x)|^2] \geq \zeta^2 \delta \exp\left\{(1 - \delta)\tilde{\vartheta}_0 K_l^{\frac{2\alpha}{\alpha+\beta(\bar{\delta}-2)}} t\right\}, \quad x \in \mathbb{R}. \quad (15)$$

2. Let us further assume that for all $r > 0$ and $x \in \mathbb{R}$

$$\min\{\rho(y : x \leq y < x + r), \rho(y : x - r < y \leq x)\} \geq C\rho(B(x, r)), \quad (16)$$

with some constant $C > 0$. If we furthermore assume that there exists a set $B \in \mathcal{B}(\mathbb{R})$ with $|B| > 0$ such that

$$\int_B u_0(x) dx > 0,$$

then for large enough fixed $T > 1$, there exists two constants $\vartheta_1 > 0, \tilde{\vartheta}_1 > 0$ independent of K_l such that

$$\inf_{x \in B(0, t^{\beta/\alpha})} \mathbb{E}[|u(t, x)|^2] \geq \vartheta_1 \delta \exp \left\{ (1 - \delta) \tilde{\vartheta}_1 K_l^{\frac{2\alpha}{\alpha + \beta(\delta - 2)}} t \right\}. \quad (17)$$

Proof: We will divide the proofs of this theorem into two steps.

Case I: Recall the mild solution of STF-SHE (1) given by (5) with $\lambda = 1$. Since the initial condition u_0 is bounded below by $\zeta > 0$, by the property of Green function $G_t(x)$, it is known that

$$\inf_{x \in \mathbb{R}} \mathbb{T}_{0,t} u_0(x) \geq \zeta.$$

Then by the assumption **(C.1)** on the coefficient $\sigma(\cdot)$ and using the Itô's isometry for the stochastic integral

$$\int_0^t \int_{\mathbb{R}} G_{t-s}(x - y) \sigma(u(s, y)) F_\rho(ds, dy), \quad t \in (0, T], \quad (18)$$

we can deduce that

$$\begin{aligned} \mathbb{E}[|u(t, x)|^2] &= (\mathbb{T}_{0,t} u_0(x))^2 + \mathbb{E} \left[\left| \int_0^t \int_{\mathbb{R}} G_{t-s}(x - y) \sigma(u(s, y)) F_\rho(ds, dy) \right|^2 \right] \\ &\geq \zeta^2 + K_l^2 \int_0^t \int_{\mathbb{R}} G_{t-s}^2(x - y) \mathbb{E}[|u(s, y)|^2] \rho(dy) ds \end{aligned} \quad (19)$$

Denote by

$$h(t) = \inf_{x \in \mathbb{R}} \mathbb{E}[|u(t, x)|^2].$$

Then let us insert the inequality (36) into (19), we have that

$$h(t) \geq \zeta^2 + C_5 K_l^2 \int_0^t h(s) \min \left\{ (t - s)^{-\frac{\beta}{\alpha}}, (t - s)^{\frac{\beta}{\alpha}(\delta - 2)} \right\} ds.$$

Now let us assume that

$$T_0 = \inf \left\{ t \in (0, T] : \min \left\{ (t - s)^{-\frac{\beta}{\alpha}}, (t - s)^{\frac{\beta}{\alpha}(\delta - 2)} \right\} \geq C_5 K_l^2 \Gamma \left(1 + \frac{\beta(\delta - 2)}{\alpha} \right) \right\}.$$

If $t \leq T_0$ then we have

$$h(t) \geq \zeta^2 + C_5 K_l^2 \min \left\{ T_0^{-\frac{\beta}{\alpha}}, T_0^{\frac{\beta}{\alpha}(\delta - 2)} \right\} \int_0^t h(s) ds,$$

Then according to Lemma 6, for any $t \in (0, T_0]$, one gets

$$h(t) \geq \zeta^2 \exp \left\{ C_5 K_l^2 \min \left\{ T_0^{-\frac{\beta}{\alpha}}, T_0^{\frac{\beta}{\alpha}(\delta - 2)} \right\} t \right\}.$$

Then we can conclude the proof of (14). We also remark that this inequality is satisfied for any fixed T_0 . So without loss of generality, we will assume that $T_0 > 1$ in this part.

If $t \geq T_0$, then also by Lemma 6, we deduce that for any $\delta \in (0, 1)$

$$h(t) \geq \zeta^2 \delta \exp \left\{ (1 - \delta) \left(C_5 K_l^2 \Gamma(1 + \frac{\beta(\underline{\delta} - 2)}{\alpha}) \min \left\{ t^{1 - \frac{\beta}{\alpha}}, t^{1 + \frac{\beta}{\alpha}(\underline{\delta} - 2)} \right\} \right)^{\frac{\alpha}{\alpha + \beta(\underline{\delta} - 2)}} + 1 \right\},$$

which implies that the (15) holds. Then the proof of the first part of this theorem is completed.

Case II: From now on, let us prove the second point of this theorem. We will follow the methods proposed in [9], [10], [24] and etc. In fact, for some $t_0 > 0$, we can rewrite the relation (19) as follows

$$\mathbb{E}[|u(t_0 + t, x)|^2] \geq (\mathbb{T}_{0, t_0+t} u_0(x))^2 + K_l^2 \int_0^t \int_{\mathbb{R}} G_{t-s_1}^2(x - y_1) \mathbb{E}[|u(t_0 + s_1, y_1)|^2] \rho(dy_1) ds_1.$$

Iterating the above inequality, we first obtain that

$$\begin{aligned} & \mathbb{E}[|u(t_0 + t, x)|^2] \\ & \geq (\mathbb{T}_{0, t_0+t} u_0(x))^2 + K_l^2 \int_0^t \int_{\mathbb{R}} G_{t-s}^2(x - y) |\mathbb{T}_{0, t_0+s} u_0(y)|^2 \rho(dy) ds \\ & + K_l^4 \int_0^t \int_{\mathbb{R}} \int_0^{s_1} \int_{\mathbb{R}} G_{t-s_1}^2(x - y_1) G_{s_1-s_2}^2(y_1 - y_2) \mathbb{E}[|u(t_0 + s_2, y_2)|^2] \rho(dy_2) ds_2 \rho(dy_1) ds_1. \end{aligned}$$

Then we can do the same calculations repeatedly and finally obtain that

$$\begin{aligned} \mathbb{E}[|u(t_0 + t, x)|^2] & \geq (\mathbb{T}_{0, t_0+t} u_0(x))^2 \\ & + \sum_{n=1}^{\infty} K_l^{2n} \int_0^t \int_{\mathbb{R}} \int_0^{s_1} \int_{\mathbb{R}} \cdots \int_0^{s_{n-1}} \int_{\mathbb{R}} \\ & \cdot |\mathbb{T}_{0, t_0+s_n} u_0(y_n)|^2 \prod_{i=1}^n G_{s_{i-1}-s_i}^2(y_{i-1} - y_i) \rho(dy_{n+1-i}) ds_{n+1-i}, \end{aligned}$$

with the convention $s_0 := t$ and $y_0 := x$.

Following Lemma 2.9 and Remark 2.10 in Foondun and Nane [10], for any fixed $t_0 > 0$ and $x \in B(0, (t_0 + t)^{\beta/\alpha})$, with $0 < s \leq t$, we have that

$$\mathbb{T}_{0, t_0+s} u_0(x) \geq g_t,$$

where the function g_t decays no faster than polynomial.

We now restrict all $y_n \in B(0, (t + t_0)^{\beta/\alpha})$ for all $n \geq 0$. We see that the value $\mathbb{E}[|u(t_0 + t, x)|^2]$ satisfies the relation

$$\begin{aligned} \mathbb{E}[|u(t_0 + t, x)|^2] & \geq g_t^2 + g_t^2 \sum_{n=1}^{\infty} K_l^{2n} \int_0^t \int_{\mathbb{R}} \int_0^{s_1} \int_{\mathbb{R}} \\ & \cdots \int_0^{s_{n-1}} \int_{B(0, (t+t_0)^{\beta/\alpha})} \prod_{i=1}^n G_{s_{i-1}-s_i}^2(y_{i-1} - y_i) \rho(dy_{n+1-i}) ds_{n+1-i}. \end{aligned} \tag{20}$$

So next we would like to establish the lower bound for the integral

$$\int_0^t \int_{\mathbb{R}} \int_0^{s_1} \int_{\mathbb{R}} \cdots \int_0^{s_{n-1}} \int_{B(0, (t+t_0)^{\beta/\alpha})} \prod_{i=1}^n G_{s_{i-1}-s_i}^2(y_{i-1} - y_i) \rho(dy_{n+1-i}) ds_{n+1-i},$$

with $n = 1, 2, \dots$ which appearing in the above (20). For each $n \in \mathbb{N}$, it is led to

$$\begin{aligned} & \int_0^t \int_{\mathbb{R}} \int_0^{s_1} \int_{\mathbb{R}} \cdots \int_0^{s_{n-1}} \int_{B(0, (t+t_0)^{\beta/\alpha})} \prod_{i=1}^n G_{s_{i-1}-s_i}^2(y_{i-1} - y_i) \rho(dy_{n+1-i}) ds_{n+1-i} \\ & \geq \int_{t-t/n}^t \int_{\mathbb{R}} \int_{s_1-t/n}^{s_1} \int_{\mathbb{R}} \cdots \int_{s_{n-1}-t/n}^{s_{n-1}} \int_{B(0, (t+t_0)^{\beta/\alpha})} \prod_{i=1}^n G_{s_{i-1}-s_i}^2(y_{i-1} - y_i) \rho(dy_{n+1-i}) ds_{n+1-i} \\ & \geq \int_0^{t/n} \int_{\mathbb{R}} \int_0^{t/n} \int_{\mathbb{R}} \cdots \int_0^{t/n} \int_{B(0, (t+t_0)^{\beta/\alpha})} \prod_{i=1}^n G_{s_i}^2(y_{i-1} - y_i) \rho(dy_{n+1-i}) ds_{n+1-i}. \end{aligned} \quad (21)$$

We will further reduce the domain of the function $\prod_{i=1}^n G_{s_i}^2(y_{i-1} - y_i)$ which has the required lower bound. For $i = 1, \dots, n$, we set

$$y_i \in \mathcal{A}_i = \left\{ y : y \in B\left(y_{i-1}, s_i^{\beta/\alpha}\right) \cap B\left(0, (t_0 + t)^{\beta/\alpha}\right) \right\},$$

with the convention $y_0 = y \in B(0, (t_0 + t)^{\beta/\alpha})$. We therefore have that $|y_i - y_{i-1}| \leq s_i^{\beta/\alpha}$. By the conditions (16) on ρ , there exists a common constant $C > 0$ such that

$$\rho(\mathcal{A}_i) \geq C \min\{s_i^{\beta/\alpha}, s_i^{\beta\delta/\alpha}\},$$

which yields that, for any $n \in \mathbb{N}$, we have

$$\begin{aligned} \int_{B(0, (t+t_0)^{\beta/\alpha})} G_{s_i}^2(y_{i-1} - y_i) \rho(dy_i) & \geq \int_{\mathcal{A}_i} G_{s_i}^2(y_{i-1} - y_i) \rho(dy_i) \\ & \geq C \min\{s_i^{-\beta/\alpha}, s_i^{\beta(\delta-2)/\alpha}\}. \end{aligned}$$

Therefore, we can easily see that the right hand of (21) can be bounded from below by

$$\begin{aligned} & \int_0^{t/n} \int_{\mathbb{R}} \int_0^{t/n} \int_{\mathbb{R}} \cdots \int_0^{t/n} \int_{\mathbb{R}} \prod_{i=1}^n G_{s_i}^2(y_{i-1} - y_i) \rho(dy_{n+1-i}) ds_{n+1-i} \\ & \geq C^n \left(\int_0^{t/n} \min\{s^{-\beta/\alpha}, s^{\beta(\delta-2)/\alpha}\} ds \right)^n \\ & \geq C^n \left(\min\{(t/n)^{1-\beta/\alpha}, (t/n)^{1+\beta(\delta-2)/\alpha}\} \right)^n. \end{aligned} \quad (22)$$

Combining the estimates (20),(21) and (22) together, one obtains that

$$\inf_{x \in B(0, (t_0+t)^{\beta/\alpha})} \mathbb{E}[|u(t_0+t, x)|^2] \geq g_t^2 \sum_{n=0}^{\infty} \left(\frac{CK_l^2 \min\{t^{1-\beta/\alpha}, t^{1+\beta(\underline{\delta}-2)/\alpha}\}}{n^{1+\beta(\underline{\delta}-2)/\alpha}} \right)^n.$$

Recall that, for any $\gamma > 0, \delta \in (0, 1)$, one has that

$$\sum_{n=0}^{\infty} \frac{k^n}{n^{\gamma n}} \geq \delta e^{1+e^{-1}(1-\delta)\gamma k^{\frac{1}{\gamma}}}.$$

Therefore for any $\delta \in (0, 1)$ and large enough t_0 , one gets

$$\begin{aligned} & \inf_{x \in B(0, (t_0+t)^{\beta/\alpha})} \mathbb{E}[|u(t_0+t, x)|^2] \\ & \geq g_t^2 \delta \exp \left\{ (1-\delta)e^{-1} (CK_l^2 \min\{t^{1-\beta/\alpha}, t^{1+\beta(\underline{\delta}-2)/\alpha}\})^{\frac{\alpha}{\alpha+\beta(\underline{\delta}-2)}} \cdot (1+\beta(\underline{\delta}-2)/\alpha) + 1 \right\}. \end{aligned}$$

Thus we can conclude the proof of (17). ■

Remark 3. Regarding the above proof of Theorem 2, the lower bound for the second moments of the exponential type for SPDE is usually more difficult than the upper bound counterpart. Such exponential type lower bound for the homogeneous noise has been obtained previously in several papers using different techniques, including the one developed by the authors in [9] and [10]. The approach used to prove the above theorem is mainly enlightened by the methods introduced in the above references, in which the localization argument is adapted. However, for the assumptions **(A)** and **(C.2)** on the intensity measure ρ , we are required to recall some renewal type inequalities which are vital to their methods, see Lemma 3.4 and Lemma 3.5 in [24] and prove some more technical estimates associated with Green function $G_t(x)$ and the measure ρ .

Let us now state the theorem related to the weakly intermittency of the solution $u(t, x)$ to STF-SHE (1), which can be immediately derived from Theorem 1 and Theorem 2. Let us firstly recall the definition of the *weak full intermittency* by introducing the upper and lower p -th moment Lyapunov exponents (see, [6], [12], [13] and etc).

Definition 3. For each $p \geq 1$, let us denote by $\overline{\mathcal{L}}(p; x)$ and respectively $\underline{\mathcal{L}}(p; x)$ the upper and lower p -th moment Lyapunov exponents of the solution $u(t, x)$ to the STF-SHE (1) at $x \in \mathbb{R}$ which is defined by

$$\overline{\mathcal{L}}(p; x) = \limsup_{t \rightarrow \infty} \frac{\log \mathbb{E}[|u(t, x)|^p]}{t}, \quad \underline{\mathcal{L}}(p; x) = \liminf_{t \rightarrow \infty} \frac{\log \mathbb{E}[|u(t, x)|^p]}{t}.$$

Then, if the smallest integer $\tilde{p}$ of $\inf_{x \in \mathbb{R}} \underline{\mathcal{L}}(p; x) > 0$ exists, that is

$$\tilde{p} = \inf \left\{ p \in \mathbb{N} : \inf_{x \in \mathbb{R}} \underline{\mathcal{L}}(p; x) > 0 \right\},$$

we say that the solution $u(t, x)$ is weakly intermittent of the order $\tilde{p}$. In particular, if $\tilde{p} = 2$, then we say that $u(t, x)$ is weakly full intermittent.

Theorem 3. *Suppose the assumptions in Theorem 1 and Theorem 2 are satisfied. Then we have that the solution $u(t, x)$ to STF-SHE (1) is weakly full intermittent. More precisely, there exist two constants $\vartheta_2, \tilde{\vartheta}_2 > 0$ independent of K_l, K_u and p such that*

$$\vartheta_2 p K_l^{\frac{2\alpha}{\alpha+\beta(\underline{\delta}-2)}} \leq p \inf_{x \in \mathbb{R}} \underline{\mathcal{L}}(2, x) \leq 2 \inf_{x \in \mathbb{R}} \underline{\mathcal{L}}(p, x) \leq 2 \sup_{x \in \mathbb{R}} \overline{\mathcal{L}}(p, x) \leq \tilde{\vartheta}_2 K_u^{\frac{2\alpha}{\alpha+\beta(\bar{\delta}-2)}} p^{1+\frac{\alpha}{\alpha+\beta(\bar{\delta}-2)}}, \quad (23)$$

holds for all $p \geq 2$.

Proof: From the definition of the weakly intermittent, it is sufficient for us to prove (23). The upper bound is an immediate result of Theorem 1. From now on, we will state the proof of the first inequality in (23), which completes the proof of this theorem by combining the non-decreasing of $\underline{\mathcal{L}}(p, x)/p$ with respect to p .

For the first case in Theorem 2, the inequality (15) yields the desired result of (23) for all $x \in \mathbb{R}$.

For the second case in Theorem 2, for any fixed x , we can choose a large enough constant $t_0 > 0$ such that $x \in B(0, t^{\beta/\alpha})$ with $t > t_0$. Therefore (17) implies that for any $t > \max\{t_0, T_0\}$

$$\mathbb{E}[|u(t, x)|^2] \geq \vartheta_1 \delta \exp \left\{ (1 - \delta) \tilde{\vartheta}_1 K_l^{\frac{2\alpha}{\alpha+\beta(\underline{\delta}-2)}} t \right\}.$$

Then we can easily obtain the lower bound in (23). As a consequence, we can conclude the proof of this theorem for both the first and the second cases. ■

As we saw, Theorem 1 and Theorem 2 make us to consider the “noise excitation” which connects the moments of the solution and the diffusion coefficient. This topic is initially investigated in [13], and then attracted many authors to study (for example, [9], [8], [10], [12], [24] and etc). More precisely, let us consider the STF-SHE (1) with the diffusive coefficient $\lambda \sigma(\cdot)$ instead of σ , where $\lambda > 0$ is called the noise intensity or the level of noise. In this case, we write $u_\lambda(t, x)$ for the solution instead of $u(t, x)$ to emphasize the noise intensity λ . In our case, we have the following results on the moment of $u_\lambda(t, x)$ as $\lambda \rightarrow \infty$.

Corollary 1. *Under the assumptions of Theorem 1 and Theorem 2, we have*

$$\frac{2\alpha}{\alpha + \beta(\underline{\delta} - 2)} \leq \liminf_{\lambda \rightarrow \infty} \frac{\log \log \mathbb{E}[|u_\lambda(t, x)|^2]}{\log \lambda} \leq \limsup_{\lambda \rightarrow \infty} \frac{\log \log \mathbb{E}[|u_\lambda(t, x)|^2]}{\log \lambda} \leq \frac{2\alpha}{\alpha + \beta(\bar{\delta} - 2)}. \quad (24)$$

Proof: According to Theorem 1, we have that for any $\lambda > 0$,

$$\sup_{x \in \mathbb{R}} \mathbb{E}[|u_\lambda(t, x)|^p] \leq L_1^{\frac{p}{2}} \left(1 + \sup_{x \in \mathbb{R}} \mathbb{E}[|u_0(x)|^p] \right) \exp \left\{ L_2 (\lambda K_u)^{\frac{2\alpha}{\alpha+\beta(\bar{\delta}-2)}} p^{1+\frac{\alpha}{\alpha+\beta(\bar{\delta}-2)}} t \right\}, \quad (25)$$

Then we can prove the right hand of (24) by taking Log of both sides of (25) and l’Hôpital’s rule.

For the lower bound, by using Theorem 2, we can replace K_l by λK_l . The for both case (I) and case (II), it is easy to obtain the left side hand of (24) for any $t > T_0$, where T_0 is the time defined in Theorem 1. However, for any $t < T_0$, we can follow the approach used in the proof of Theorem 1.4 in [9] to prove our result. We omitted the proof here. ■

Remark 4. *Following the definition firstly given in [13], we denote by the energy of the solution as follows:*

$$\mathcal{E}_t(\lambda) = \sqrt{\int_{\mathbb{R}} \mathbb{E}|u_\lambda(t, x)|^2 dx}.$$

We also recall the lower and respectively the upper nonlinear excitation index of $u_\lambda(t, x)$ at time t as follows:

$$\underline{e}(t) = \liminf_{\lambda \rightarrow \infty} \frac{\log \log \mathcal{E}_t(\lambda)}{\log \lambda}, \quad \bar{e}(t) = \limsup_{\lambda \rightarrow \infty} \frac{\log \log \mathcal{E}_t(\lambda)}{\log \lambda}.$$

If the energy of the solution exists, that is $\mathcal{E}_t(\lambda) < +\infty$, then, from Corollary 1, we can obtain that, for all $t > 0$,

$$\frac{2\alpha}{\alpha + \beta(\underline{\delta} - 2)} \leq \underline{e}(t) \leq \bar{e}(t) \leq \frac{2\alpha}{\alpha + \beta(\bar{\delta} - 2)}.$$

In particular, if the catalyst ρ is absolutely continuous with respect to the Lebesgue measure, (i.e. $\rho(dx) = cdx$), then one obtains that $\bar{\delta} = \underline{\delta} = 1$. This means excitation index $e(t)$ of $u_\lambda(t, x)$ at time t

$$e(t) = \lim_{\lambda \rightarrow \infty} \frac{\log \log \mathcal{E}_t(\lambda)}{\log \lambda} = \frac{2\alpha}{\alpha - \beta},$$

which extends the result obtained in Theorem 1.8 in [10].

Appendix: Technical lemmas

In this section, some technical lemmas are established. They played an important role in the proofs of main theorems stated in the above sections.

The following Lemma 5 comes from Lemma 3.1 and Lemma 3.3 in [24] which are stated as follows.

Lemma 5. *1. Let the intensity measure ρ satisfy the assumption (A.1) and (A.2). Then there exists a constant $C_{11} > 0$ such that for all $r > 0$,*

$$\sup_{x \in \mathbb{R}} \int_{\mathbb{R}} \exp \left\{ -\frac{(x-y)^2}{r} \right\} \rho(dy) \leq C_{11} \max \{ r^{\frac{1}{2}}, r^{\frac{\bar{\delta}}{2}} \}. \quad (26)$$

2. Suppose that the intensity measure ρ satisfies the assumption (C.2). Then the following statements are satisfied.

(a) There exists a constant $C_{12} > 0$ such that for all $r \in (0, 1]$

$$\inf_{x \in \mathbb{R}} \int_{\mathbb{R}} \exp \left\{ -\frac{(x-y)^2}{r} \right\} \rho(dy) \geq C_{12} r^{\frac{1}{2}\delta}, \quad (27)$$

and then we have that there exists a constant $C_{13} > 0$ such that for any $r > 0$,

$$\inf_{x \in \mathbb{R}} \int_{\mathbb{R}} \exp \left\{ -\frac{(x-y)^2}{r} \right\} \rho(dy) \geq C_{13} e^{-1} \min\{r^{\frac{1}{2}\delta}, r^{\frac{1}{2}}\}. \quad (28)$$

(b) The following potential type relation holds

$$\inf_{x \in \mathbb{R}} \int_{\mathbb{R}} |x-y|^{-\delta} \rho(dy) > C_l, \quad (29)$$

where $C_l > 0$ is the same constant as that in the assumption **(C.2)**.

The following Lemma 6 comes from Lemma 3.5 in [24] which is also stated as follows for the readers' convenience.

Lemma 6. Assume that $0 < \gamma_1 < \gamma_2 < 1$ and the function $h(t)$ is non-negative and locally integrable on $[0, \infty)$. Furthermore, we assume there exist two constants $\kappa_0, \kappa_2 > 0$ such that for $t > 0$

$$h(t) \geq \kappa_0 + \kappa_1 \int_0^t \min\{(t-s)^{\gamma_1-1}, (t-s)^{\gamma_2-1}\} h(s) ds.$$

Then for any t satisfying $\min\{t^{\gamma_1}, t^{\gamma_2}\} \geq (\kappa_1 \Gamma(\gamma))^{-1}$, we have that

$$h(t) \geq \frac{\kappa_0}{(\kappa_1 \Gamma(\gamma_2) \min\{t^{\gamma_1}, t^{\gamma_2}\})^{\frac{1}{\gamma_2}}} \exp \left\{ (\kappa_1 \Gamma(\gamma_2) \min\{t^{\gamma_1}, t^{\gamma_2}\})^{\frac{1}{\gamma_2}} \right\}.$$

Furthermore, for any $\delta \in (0, 1)$,

$$h(t) \geq \kappa_0 \delta \exp \left\{ (1-\delta)(\kappa_1 \Gamma(\gamma_2) \min\{t^{\gamma_1}, t^{\gamma_2}\})^{\frac{1}{\gamma_2}} + 1 \right\}.$$

We also need recall the following lemma which has been proved in Foondun and Nane [10] in the multidimensional case (i.e. $d \geq 1$). But in our case we just need the bounds (30) and (31) with $d = 1$.

Lemma 7. 1. There exists a positive constant C_1 such that for all $x \in \mathbb{R}^d$ with $d \geq 1$

$$G_t(x) \geq C_1 \left(t^{-\beta d/\alpha} \wedge \frac{t^\beta}{|x|^{d+\alpha}} \right). \quad (30)$$

2. If we further suppose that $\alpha > d$, there exists a positive constant C_2 such that for all $x \in \mathbb{R}^d$

$$G_t(x) \leq C_2 \left(t^{-\beta d/\alpha} \wedge \frac{t^\beta}{|x|^{d+\alpha}} \right). \quad (31)$$

Based on the above Lemma 7, especially the upper bound (31) for the Green function $G_t(x)$, one can prove the following

Lemma 8. *Let the intensity measure ρ satisfy the assumption (A) and assume that $1 < \alpha \leq 2$, for any $0 \leq t \leq T$, we have that*

$$\sup_{x \in \mathbb{R}} \int_{\mathbb{R}} G_t(x-y) \rho(dy) \leq C \max\{1, t^{\frac{\beta(\bar{\delta}-1)}{\alpha}}\}, \quad (32)$$

for some constant $C > 0$ independent of t and x .

Proof: From (31) with $d = 1$, we can write that

$$\int_{\mathbb{R}} G_t(x-y) \rho(dy) \leq C(\mathcal{I}_1 + \mathcal{I}_2),$$

with

$$\begin{aligned} \mathcal{I}_1 &= t^{-\frac{\beta}{\alpha}} \int_{\mathbb{R}} 1_{\{|x-y|^{\alpha+1} \leq t^{\beta(1+\frac{1}{\alpha})}\}}(y) \rho(dy) \\ \mathcal{I}_2 &= t^\beta \int_{\mathbb{R}} \frac{1}{|x-y|^{1+\alpha}} 1_{\{|x-y|^{\alpha+1} > t^{\beta(1+\frac{1}{\alpha})}\}}(y) \rho(dy). \end{aligned}$$

For the first term $\mathcal{I}_1$, we have

$$\begin{aligned} e^{-1} \int_{\mathbb{R}} 1_{\{|x-y|^{\alpha+1} \leq t^{\beta(1+\frac{1}{\alpha})}\}}(y) \rho(dy) &\leq \int_{\mathbb{R}} \exp\left\{-\frac{|x-y|^2}{t^{\frac{2\beta}{\alpha}}}\right\} \rho(dy) \\ &\leq C \max\left\{t^{\beta/\alpha}, t^{\beta\bar{\delta}/\alpha}\right\}, \end{aligned}$$

which implies that

$$\mathcal{I}_1 \leq C \max\{1, t^{\beta(\bar{\delta}-1)/\alpha}\}.$$

For the second term $\mathcal{I}_2$, we have

$$\begin{aligned} \int_{\mathbb{R}} \frac{1}{|x-y|^{1+\alpha}} 1_{\{|x-y| > t^{\frac{\beta}{\alpha}}\}}(y) \rho(dy) &= \int_0^\infty \rho\left(y : \frac{1}{|x-y|^{1+\alpha}} 1_{\{|x-y| > t^{\frac{\beta}{\alpha}}\}} \geq u\right) du \\ &= \int_0^{t^{-\frac{\beta}{\alpha}(1+\alpha)}} \rho(B(x, u^{-\frac{1}{1+\alpha}})) du \\ &\leq C \int_0^{t^{-\frac{\beta}{\alpha}(1+\alpha)}} u^{-\frac{\bar{\delta}}{1+\alpha}} du \\ &= C \frac{1+\alpha-\bar{\delta}}{1+\alpha} t^{\frac{\beta(\bar{\delta}-1)}{\alpha}-\beta}, \end{aligned}$$

which implies that

$$\mathcal{I}_2 \leq C \frac{1 + \alpha - \bar{\delta}}{1 + \alpha} t^{\frac{\beta(\bar{\delta}-1)}{\alpha}}.$$

Then by combining the above estimates for I_1 and I_2 , we can conclude the proof of this lemma. $\blacksquare$

Lemma 9. *Let the intensity measure ρ satisfy the assumption (A.1) and (A.2), for any $\theta > 0$, $1 < \alpha \leq 2$ and $\alpha + \beta(\bar{\delta} - 2) > 0$, we have that*

$$\sup_{t \in [0, T]} \sup_{x \in \mathbb{R}} \int_0^t \int_{\mathbb{R}} e^{-\theta s} G_s^2(x - y) \rho(dy) ds \leq C_3 C_{\bar{\delta}}^*(\alpha, \beta, \theta), \quad (33)$$

for some constant $C_3 > 0$ and $C_{\bar{\delta}}^*(\alpha, \beta, \theta)$ is defined as follows

$$\begin{aligned} C_{\bar{\delta}}^*(\alpha, \beta, \theta) = & \left(\Gamma(1 - \beta/\alpha) \theta^{\beta/\alpha - 1} + \Gamma(1 + \beta(\bar{\delta} - 2)/\alpha) \theta^{-1 + \frac{\beta(2 - \bar{\delta})}{\alpha}} \right) \\ & + \Gamma \left(1 + 2\beta - \frac{\beta(2\alpha + 2 - \bar{\delta})}{\alpha} \right) \theta^{-1 - 2\beta + \frac{\beta(2\alpha + 2 - \bar{\delta})}{\alpha}}. \end{aligned} \quad (34)$$

.

Proof: From (31) with $d = 1$, we have

$$\int_0^t \int_{\mathbb{R}} e^{-\theta s} G_s^2(x - y) \rho(dy) ds \leq C(\mathcal{J}_1 + \mathcal{J}_2),$$

with

$$\begin{aligned} \mathcal{J}_1 &= \int_0^t e^{-\theta s} s^{-\frac{2\beta}{\alpha}} \int_{\mathbb{R}} 1_{\{|x-y| \leq s^{\frac{\beta}{\alpha}}\}} \rho(dy) ds \\ \mathcal{J}_2 &= \int_0^t e^{-\theta s} s^{2\beta} \int_{\mathbb{R}} \frac{1}{|x-y|^{2(1+\alpha)}} 1_{\{|x-y| > s^{\frac{\beta}{\alpha}}\}} \rho(dy) ds. \end{aligned}$$

Then for the first term $\mathcal{J}_1$, by using the similar arguments in the proof of Lemma 8, we have

$$\begin{aligned} \mathcal{J}_1 &\leq C \int_0^t e^{-\theta s} s^{-\frac{2\beta}{\alpha}} \max \left\{ s^{\beta/\alpha}, s^{\beta\bar{\delta}/\alpha} \right\} ds \\ &\leq C \int_0^\infty e^{-\theta s} (s^{-\beta/\alpha} + s^{\beta(\bar{\delta}-2)/\alpha}) ds \\ &= C \left(\Gamma(1 - \beta/\alpha) \theta^{\beta/\alpha - 1} + \Gamma(1 + \beta(\bar{\delta} - 2)/\alpha) \theta^{-1 + \frac{\beta(2 - \bar{\delta})}{\alpha}} \right). \end{aligned}$$

Now let us move to the second term J_2 . Firstly, one gets

$$\begin{aligned} \int_{\mathbb{R}} \frac{1}{|x-y|^{2(1+\alpha)}} 1_{\{|x-y|>s^{\beta/\alpha}\}}(y) \rho(dy) &= \int_0^\infty \rho\left(y : \frac{1}{|x-y|^{2(1+\alpha)}} 1_{\{|x-y|>s^{\beta/\alpha}\}} \geq u\right) du \\ &= \int_0^{s^{-\frac{2\beta(1+\alpha)}{\alpha}}} \rho(B(x, u^{-\frac{1}{2(\alpha+1)}})) du \\ &\leq C s^{-\frac{\beta(2\alpha+2-\bar{\delta})}{\alpha}}. \end{aligned}$$

Then we have

$$\begin{aligned} \mathcal{J}_2 &\leq C \int_0^\infty e^{-\theta s} s^{-\frac{\beta(2\alpha+2-\bar{\delta})}{\alpha}+2\beta} ds \\ &\leq C \Gamma\left(1+2\beta-\frac{\beta(2\alpha+2-\bar{\delta})}{\alpha}\right) \theta^{-1-2\beta+\frac{\beta(2\alpha+2-\bar{\delta})}{\alpha}}, \end{aligned}$$

because $\alpha + \beta(\bar{\delta} - 2) > 0$. This concludes the proof of this lemma. ■

The following lemma plays an important role in the proof of the lower bound of Lyapunov exponent $\underline{\mathcal{L}}(p, x)$.

Lemma 10. *Suppose that $0 < \alpha \leq 2$ and the intensity measure ρ satisfies the assumption (C.2). Then we have that*

$$\inf_{x \in \mathbb{R}} \int_{\mathbb{R}} G_t(x-y) \rho(dy) \geq C_4 \min\{1, t^{\frac{\beta}{\alpha}(\bar{\delta}-1)}\}, \quad (35)$$

and

$$\inf_{x \in \mathbb{R}} \int_{\mathbb{R}} G_t^2(x-y) \rho(dy) \geq C_5 \min\left\{t^{-\frac{\beta}{\alpha}}, t^{\frac{\beta}{\alpha}(\bar{\delta}-2)}\right\}, \quad (36)$$

hold for some constants $C_4 > 0, C_5 > 0$ independent of t .

Proof: Recall the lower bound (30) for the Green function $G_t(x-y)$, we can write that

$$\int_{\mathbb{R}} G_t(x-y) \rho(dy) \geq C \int_{\mathbb{R}} \left[t^{-\frac{\beta}{\alpha}} 1_{\{|x-y| \leq t^{\frac{\beta}{\alpha}}\}} + \frac{t^\beta}{|x-y|^{1+\alpha}} 1_{\{|x-y| > t^{\frac{\beta}{\alpha}}\}} \right] \rho(dy). \quad (37)$$

Then by using (28) and the assumption (C.2) for ρ , we have

$$\begin{aligned}
& \int_{\mathbb{R}} \left[t^{-\frac{\beta}{\alpha}} 1_{\{|x-y| \leq t^{\frac{\beta}{\alpha}}\}} + \frac{t^{\beta}}{|x-y|^{1+\alpha}} 1_{\{|x-y| > t^{\frac{\beta}{\alpha}}\}} \right] \rho(dy) \\
& \geq t^{-\frac{\beta}{\alpha}} \int_{\mathbb{R}} \exp \left\{ -\frac{|x-y|^2}{t^{\frac{2\beta}{\alpha}}} \right\} 1_{\{|x-y| \leq t^{\frac{\beta}{\alpha}}\}} \rho(dy) \\
& \quad + t^{\beta} \int_{\mathbb{R}} \frac{1}{|x-y|^{1+\alpha}} 1_{\{|x-y| > t^{\frac{\beta}{\alpha}}\}} \rho(dy) \\
& \geq Ce^{-1} t^{-\frac{\beta}{\alpha}} \min\{t^{\frac{\beta}{\alpha}}, t^{\frac{\beta}{\alpha}\delta}\} \\
& \quad + t^{\beta} \int_0^{+\infty} \rho \left(y : \frac{1}{|x-y|^{1+\alpha}} 1_{\{|x-y| > t^{\frac{\beta}{\alpha}}\}} \geq u \right) du \\
& = Ce^{-1} \min\{1, t^{\frac{\beta}{\alpha}(\delta-1)}\} + Ct^{\beta} \int_0^{t^{-\frac{\beta}{\alpha}(1+\alpha)}} \rho(B(x, u^{-\frac{1}{1+\alpha}})) du \\
& \geq Ce^{-1} \min\{1, t^{\frac{\beta}{\alpha}(\delta-1)}\} + Ct^{\beta} \int_0^{t^{-\frac{\beta}{\alpha}(1+\alpha)}} u^{-\frac{\delta}{1+\alpha}} du \\
& = Ce^{-1} \min\{1, t^{\frac{\beta}{\alpha}(\delta-1)}\} + Ct^{\frac{\beta}{\alpha}(\delta-1)},
\end{aligned} \tag{38}$$

which yields the inequality (35).

Now let us move to the proof of (36). In fact, we have that

$$\begin{aligned}
\int_{\mathbb{R}} G_t^2(x-y) \rho(dy) &= C \int_{\mathbb{R}} \left[t^{-\frac{\beta}{\alpha}} 1_{\{|x-y| \leq t^{\frac{\beta}{\alpha}}\}} + \frac{t^{\beta}}{|x-y|^{1+\alpha}} 1_{\{|x-y| > t^{\frac{\beta}{\alpha}}\}} \right]^2 \rho(dy) \\
&\geq Ct^{-\frac{2\beta}{\alpha}} \int_{\mathbb{R}} 1_{\{|x-y| \leq t^{\frac{\beta}{\alpha}}\}} \rho(dy) \\
&\quad + Ct^{2\beta} \int_{\mathbb{R}} \frac{1}{|x-y|^{2(1+\alpha)}} 1_{\{|x-y| > t^{\frac{\beta}{\alpha}}\}} \rho(dy) \\
&:= C(\mathcal{K}_1 + \mathcal{K}_2).
\end{aligned} \tag{39}$$

One can use the similar arguments in the proof of (35). Then we have that

$$\mathcal{K}_1 \geq t^{-\frac{2\beta}{\alpha}} \min\{t^{\frac{\beta}{\alpha}}, t^{\frac{\beta}{\alpha}\delta}\} = \min\{t^{-\frac{\beta}{\alpha}}, t^{\frac{\beta}{\alpha}(\delta-2)}\}$$

Also using the similar arguments in (35), one has that

$$\begin{aligned}
\mathcal{K}_2 &= t^{2\beta} \int_{\mathbb{R}} \frac{1}{|x-y|^{2(1+\alpha)}} 1_{\{|x-y| > t^{\frac{\beta}{\alpha}}\}} \rho(dy) \\
&\geq t^{2\beta} \int_0^{+\infty} \rho\left(y : \frac{1}{|x-y|^{2(1+\alpha)}} 1_{\{|x-y| > t^{\frac{1}{\alpha}}\}} \geq u\right) du \\
&= t^{2\beta} \int_0^{t^{-\frac{2\beta}{\alpha}(1+\alpha)}} u^{-\frac{1}{2(1+\alpha)}\delta} du \\
&= \frac{2(1+\alpha)}{2+2\alpha-\delta} t^{\frac{\beta}{\alpha}(\delta-2)}.
\end{aligned} \tag{40}$$

Then combining the estimates for $\mathcal{K}_1$ and $\mathcal{K}_2$ defined in (39), one can conclude the proof of (36) where C_5 is positive a constant which does not depend on t and x . ■

The following lemma plays an important role in the proof of Lemma 4 which concerns the moment estimate for the increments of $\mathbb{S}u(t, x)$ with respect to the time and space variables.

Lemma 11. *Let ρ satisfy the assumption **(A)** and assume that $1 < \alpha \leq 2$ and $0 < \beta < \min\{1, \frac{\alpha}{2-\delta}\}$. Then for each fixed $T > 0$, there exists a positive constant C such that for all $0 < t_1 < t_2 \leq T$ and $x_1, x_2 \in \mathbb{R}$,*

$$\int_0^{t_2} \int_{\mathbb{R}} |G_{t_2-s}(x_2 - y) - G_{t_1-s}(x_1 - y)|^2 \rho(dy) ds \leq C (|t_2 - t_1|^{2\mu} + |x_2 - x_1|^{2\kappa}), \tag{41}$$

with $0 < \mu < \frac{\alpha+\beta(\delta-2)}{2\alpha}$ and $0 < \kappa < \frac{\alpha+\beta(\delta-2)}{2\beta}$.

Proof: Let $0 < t_1 < t_2 \leq T$ and $x_1, x_2 \in \mathbb{R}$, we can write that

$$\int_0^{t_2} \int_{\mathbb{R}} |G_{t_2-s}(x_2 - y) - G_{t_1-s}(x_1 - y)|^2 \rho(dy) ds \leq 2(\mathcal{L} + \mathcal{M} + \mathcal{N}),$$

where we denote by

$$\begin{aligned}
\mathcal{L} &= \int_{t_1}^{t_2} \int_{\mathbb{R}} |G_{t_2-s}(x_2 - y)|^2 \rho(dy) ds \\
\mathcal{M} &= \int_0^{t_1} \int_{\mathbb{R}} |G_{t_2-s}(x_1 - y) - G_{t_1-s}(x_1 - y)|^2 \rho(dy) ds \\
\mathcal{N} &= \int_0^{t_2} \int_{\mathbb{R}} |G_{t_2-s}(x_2 - y) - G_{t_2-s}(x_1 - y)|^2 \rho(dy) ds.
\end{aligned}$$

Now let us establish proper upper bounds for $\mathcal{L}, \mathcal{M}, \mathcal{N}$ respectively.

For the first term $\mathcal{L}$, by using the upper bound (31) for $G_t(x)$ with $d = 1$ and Lemma 8, we have

$$\begin{aligned}
\mathcal{L} &\leq C \int_{t_1}^{t_2} \left((t_2 - s)^{-\frac{2\beta}{\alpha}} \int_{\mathbb{R}} 1_{\{|x-y| \leq (t_2-s)^{\frac{\beta}{\alpha}}\}} \rho(dy) \right. \\
&\quad \left. + (t_2 - s)^{2\beta} \int_{\mathbb{R}} \frac{1}{|x-y|^{2(1+\alpha)}} 1_{\{|x-y| > (t_2-s)^{\frac{\beta}{\alpha}}\}} \rho(dy) \right) ds \\
&\leq C \int_{t_1}^{t_2} \left((t_2 - s)^{-\frac{\beta}{\alpha}} + (t_2 - s)^{\frac{\beta(\bar{\delta}-2)}{\alpha}} + (t_2 - s)^{2\beta - \frac{\beta(2\alpha+2-\bar{\delta})}{\alpha}} \right) ds \\
&= C \left((t_2 - t_1)^{1-\frac{\beta}{\alpha}} + (t_2 - t_1)^{1+\frac{\beta(\bar{\delta}-2)}{\alpha}} + (t_2 - t_1)^{1+2\beta - \frac{\beta(2\alpha+2-\bar{\delta})}{\alpha}} \right) \\
&= C(t_2 - t_1)^{\frac{\alpha+\beta(\bar{\delta}-2)}{\alpha}} \left[(t_2 - t_1)^{\frac{\beta(1-\bar{\delta})}{\alpha}} + 1 \right] \\
&\leq C(t_2 - t_1)^{\frac{\alpha+\beta(\bar{\delta}-2)}{\alpha}} \left[T^{\frac{\beta(1-\bar{\delta})}{\alpha}} + 1 \right] \\
&:= C(t_2 - t_1)^{\frac{\alpha+\beta(\bar{\delta}-2)}{\alpha}},
\end{aligned} \tag{42}$$

because the exponential $\frac{\beta(1-\bar{\delta})}{\alpha}$ is positive.

Now let us move to the second term $\mathcal{M}$. One obtains that

$$\begin{aligned}
\mathcal{M} &= \int_0^{t_1} \int_{\mathbb{R}} |G_{t_2-s}(x_1 - y) - G_{t_1-s}(x_1 - y)|^2 \rho(dy) ds \\
&= \int_0^{t_1-|t_1-t_2|} \int_{\mathbb{R}} |G_{t_2-s}(x_1 - y) - G_{t_1-s}(x_1 - y)|^2 \rho(dy) ds \\
&\quad + \int_{t_1-|t_1-t_2|}^{t_1} \int_{\mathbb{R}} |G_{t_2-s}(x_1 - y) - G_{t_1-s}(x_1 - y)|^2 \rho(dy) ds \\
&:= \mathcal{M}_1 + \mathcal{M}_2.
\end{aligned} \tag{43}$$

The second term $\mathcal{M}_2$ in (43) can be bounded as follows

$$\mathcal{M}_2 \leq \int_{t_1-|t_1-t_2|}^{t_1} \int_{\mathbb{R}} (G_{t_2-s}^2(x_1 - y) + G_{t_1-s}^2(x_1 - y) + 2G_{t_2-s}(x_1 - y)G_{t_1-s}(x_1 - y)) \rho(dy) ds.$$

Following the similar arguments in the proof of (42), we have that

$$\mathcal{M}_2 \leq C(t_2 - t_1)^{\frac{\alpha+\beta(\bar{\delta}-2)}{\alpha}}. \tag{44}$$

Now let us consider the first term $\mathcal{M}_1$ in (43). Let us first recall some facts on the Green function $G_t(x)$. Denote by X_t a symmetric α -stable process with density function $p(t, x)$. This is characterized through the Fourier transform which is given by

$$p^*(t, \xi) = e^{-t\nu|\xi|^\alpha}.$$

Let $D = \{D_r, r \geq 0\}$ denote a ϱ -stable subordinator and E_t be its first passage time. It is known that the density of the time changed process X_{E_t} is given by $G_t(x)$. By conditioning, we have the Green function $G_t(x)$ can be written as (see, for example, [10])

$$G_t(x) = \int_0^\infty p(s, x) f_{E_t}(s) ds,$$

where $f_{E_t}(x) = t\beta^{-1}x^{-1-1/\beta}g_\beta(tx^{-1/\beta})$, where $g_\beta(\cdot)$ is the density function of D_1 and is infinitely differentiable on the entire real line, with $g_\beta(u) = 0$ if $u \leq 0$. Moreover,

$$g_\beta(u) \sim K(\beta u)^{\frac{1-\beta/2}{1-\beta}} \exp\{-|1-\beta|(u/\beta)^{\beta/(\beta-1)}\}, \quad \text{as } u \rightarrow 0+,$$

and

$$g_\beta(u) \sim \frac{\beta}{\Gamma(1-\beta)} u^{-1-\beta}, \quad \text{as } u \rightarrow \infty.$$

We also need the following properties about the heat kernel of stable process (see, for example, [10]).

$$p(t, x) = t^{-\frac{1}{\alpha}} p(1, t^{-1/\alpha} x), \quad p(st, x) = s^{-1/\alpha} p(t, s^{-1/\alpha} x),$$

and (see, for example, Proposition 2.6 in [14]).

$$\left| \frac{\partial}{\partial t} p(t, x) \right| \leq C \frac{p(t, x)}{t},$$

and

$$p(t, x - y) \leq C \left[t^{-1/\alpha} 1_{\{t \geq |x-y|^\alpha\}} + \frac{t}{|x-y|^{1+\alpha}} 1_{\{t < |x-y|^\alpha\}} \right]. \quad (45)$$

Thus we have that

$$\begin{aligned} & G_{t_2-s}(x_2 - y) - G_{t_1-s}(x_2 - y) \\ & \leq \int_0^\infty \left| p\left(\left(\frac{t_2-s}{u}\right)^\beta, x-y\right) - p\left(\left(\frac{t_1-s}{u}\right)^\beta, x-y\right) \right| g_\beta(u) du \\ & \leq \int_0^\infty \int_{\left(\frac{t_1-s}{u}\right)^\beta}^{\left(\frac{t_2-s}{u}\right)^\beta} \left| \frac{\partial}{\partial v} p(v, x-y) \right| dv g_\beta(u) du \\ & \leq C \int_0^\infty \int_{\left(\frac{t_1-s}{u}\right)^\beta}^{\left(\frac{t_2-s}{u}\right)^\beta} \frac{1}{v} p(v, x-y) dv g_\beta(u) du \\ & \leq C \int_0^\infty \int_{\left(\frac{t_1-s}{u}\right)^\beta}^{\left(\frac{t_2-s}{u}\right)^\beta} \frac{1}{v} \left[v^{-1/\alpha} 1_{\{v \geq |x-y|^\alpha\}} + \frac{v}{|x-y|^{1+\alpha}} 1_{\{v < |x-y|^\alpha\}} \right] dv g_\beta(u) du \\ & \leq C \int_0^\infty \left[\left(\left(\frac{t_1-s}{u}\right)^{-\beta/\alpha} - \left(\frac{t_2-s}{u}\right)^{-\beta/\alpha} \right) 1_{\{(\frac{t_2-s}{u})^\beta \geq |x-y|^\alpha\}} \right. \\ & \quad \left. + \frac{1}{|x-y|^{1+\alpha}} (|t_2-s|^\beta - |t_1-s|^\beta) u^{-\beta} 1_{\{(\frac{t_2-s}{u})^\beta < |x-y|^\alpha\}} \right] g_\beta(u) du. \end{aligned} \quad (46)$$

Suppose that $|x - y| \leq (t_2 - s)^{\beta/\alpha}$, then $(t_2 - s)/|x - y|^{\alpha/\beta} \geq 1$. For $u \leq \frac{t_2 - s}{|x - y|^{\alpha/\beta}}$, we can write the first term in the right hand of the inequality (46) that

$$\begin{aligned} & [(t_1 - s)^{-\beta/\alpha} - (t_2 - s)^{-\beta/\alpha}] \int_0^\infty u^{\beta/\alpha} g_\beta(u) 1_{\left\{u \leq \frac{t_2 - s}{|x - y|^{\alpha/\beta}}\right\}} du \\ & \leq C [(t_1 - s)^{-\beta/\alpha} - (t_2 - s)^{-\beta/\alpha}] \left[\int_0^1 u^{\beta/\alpha} g_\beta(u) du + \int_1^{\frac{t_2 - s}{|x - y|^{\alpha/\beta}}} u^{\beta/\alpha} g_\beta(u) du \right], \end{aligned}$$

where we have used the fact that

$$\int_0^1 u^{\beta/\alpha} g_\beta(u) du + \int_1^{\frac{t_2 - s}{|x - y|^{\alpha/\beta}}} u^{\beta/\alpha} g_\beta(u) du < \infty,$$

because of the asymptotic behavior of $g_\beta(u)$ as $u \rightarrow \infty$ and $u \rightarrow 0+$. On the other hand, for the second term in the last inequality also with $|x - y| \leq (t_2 - s)^{\beta/\alpha}$, we have that

$$\begin{aligned} & \frac{1}{|x - y|^{1+\alpha}} (|t_2 - s|^\beta - |t_1 - s|^\beta) \int_0^\infty u^{-\beta} 1_{\left\{u > \frac{t_2 - s}{|x - y|^{\alpha/\beta}}\right\}} g_\beta(u) du \\ & \leq \frac{1}{|x - y|^{1+\alpha}} (|t_2 - s|^\beta - |t_1 - s|^\beta) \int_1^\infty u^{-\beta} g_\beta(u) du. \end{aligned}$$

The integral $\int_1^\infty u^{-\beta} g_\beta(u) du$ is finite due to the asymptotic behavior of $g_\beta(u)$ as $u \rightarrow \infty$.

Suppose now that $|x - y| > (t_2 - s)^{\beta/\alpha}$, then $(t_2 - s)/|x - y|^{\alpha/\beta} < 1$. When we have $u \leq \frac{t_2 - s}{|x - y|^{\alpha/\beta}} < 1$, following the similar arguments in the above case $|x - y| \leq (t_2 - s)^{\beta/\alpha}$, we claim that the integral $\int_0^\infty u^{\beta/\alpha} g_\beta(u) 1_{\left\{u \leq \frac{t_2 - s}{|x - y|^{\alpha/\beta}}\right\}} du$ can be bounded by $\int_0^1 u^{\beta/\alpha} g_\beta(u) du$ which is finite.

Similarly, we also have that the integral

$$\int_0^\infty u^{-\beta} 1_{\left\{u > \frac{t_2 - s}{|x - y|^{\alpha/\beta}}\right\}} g_\beta(u) du \leq \int_0^\infty u^{-\beta} g_\beta(u) du < \infty,$$

because of the asymptotic behavior of $g_\beta(u)$ as $u \rightarrow \infty$ and $u \rightarrow 0+$. Combining the above arguments, we conclude that

$$\begin{aligned} |G_{t_2 - s}(x - y) - G_{t_1 - s}(x - y)| & \leq C \left[|(t_1 - s)^{-\beta/\alpha} - (t_2 - s)^{-\beta/\alpha}| 1_{\{|x - y| \leq (t_2 - s)^{\beta/\alpha}\}} \right. \\ & \quad \left. + \frac{1}{|x - y|^{1+\alpha}} ||t_2 - s|^\beta - |t_1 - s|^\beta| 1_{\{|x - y| > (t_2 - s)^{\beta/\alpha}\}} \right]. \end{aligned} \quad (47)$$

So with Lemma 8, one obtains that

$$\begin{aligned}
\mathcal{M}_1 &= \int_0^{t_1-|t_1-t_2|} \int_{\mathbb{R}} |G_{t_2-s}(x_1-y) - G_{t_1-s}(x_1-y)|^2 \rho(dy) ds \\
&\leq C \int_0^{t_1-|t_1-t_2|} ((t_1-s)^{-\beta/\alpha} - (t_2-s)^{-\beta/\alpha})^2 \int_{\mathbb{R}} 1_{\{|x-y| \leq (t_2-s)^{\beta/\alpha}\}} \rho(dy) ds \\
&\quad + C \int_0^{t_1-|t_1-t_2|} (|t_2-s|^\beta - |t_1-s|^\beta)^2 \int_{\mathbb{R}} \frac{1}{|x-y|^{2(1+\alpha)}} 1_{\{|x-y| > (t_2-s)^{\beta/\alpha}\}} \rho(dy) ds \\
&\leq C \int_0^{t_1-|t_1-t_2|} \max \left\{ (t_1-s)^{-2\beta/\alpha} (t_2-s)^{\beta/\alpha} - (t_2-s)^{-\beta/\alpha}, \right. \\
&\quad \left. (t_1-s)^{-2\beta/\alpha} (t_2-s)^{\beta\bar{\delta}/\alpha} - (t_2-s)^{\beta(\bar{\delta}-2)/\alpha} \right\} ds \\
&\quad + C \int_0^{t_1-|t_1-t_2|} \left[(t_2-s)^{\frac{\beta(4\alpha+2-\bar{\delta})}{\alpha}} + (t_1-s)^{2\beta} (t_2-s)^{\frac{\beta(2\alpha+2-\bar{\delta})}{\alpha}} \right] ds \\
&\leq C(t_2-t_1)^{\frac{\alpha+\beta(\bar{\delta}-2)}{\alpha}}.
\end{aligned} \tag{48}$$

The above estimates for $\mathcal{M}_1$ and $\mathcal{M}_2$ imply that

$$\mathcal{M} \leq C(t_2-t_1)^{\frac{\alpha+\beta(\bar{\delta}-2)}{\alpha}}.$$

Now let us prove the last term $\mathcal{N}$. Recall the proofs of Proposition 2.1 in [1], for some constant κ such that $0 < \kappa < \alpha$, we can write

$$\begin{aligned}
&|G_{t_2-s}(x_2-y) - G_{t_2-s}(x_1-y)| \\
&\leq \int_0^\infty |p(((t_2-s)/u)^\beta, x_2-y) - p(((t_2-s)/u)^\beta, x_1-y)| g_\beta(u) du \\
&\leq C \frac{|x_2-x_1|^\kappa}{(t_2-s)^{\frac{\kappa\beta}{\alpha}}} \int_0^\infty u^{\frac{\kappa\beta}{\alpha}} \left[p\left(\left((t_2-s)/u\right)^\beta, \frac{x_2-y}{2}\right) + p\left(\left((t_2-s)/u\right)^\beta, \frac{x_1-y}{2}\right) \right] g_\beta(u) du.
\end{aligned}$$

Then we can firstly estimate the inner integral in $\mathcal{N}$ as follows

$$\begin{aligned}
& \int_{\mathbb{R}} |G_{t_2-s}(x_2-y) - G_{t_2-s}(x_1-y)|^2 \rho(dy) \\
& \leq C \frac{|x_2-x_1|^{2\kappa}}{(t_2-s)^{\frac{2\kappa\beta}{\alpha}}} \int_{\mathbb{R}} \left| \int_0^\infty u^{\frac{\kappa\beta}{\alpha}} \left[p\left(\left((t_2-s)/u\right)^\beta, \frac{x_2-y}{2}\right) \right. \right. \\
& \quad \left. \left. + p\left(\left((t_2-s)/u\right)^\beta, \frac{x_1-y}{2}\right) \right] g_\beta(u) du \right|^2 \rho(dy) \\
& \leq C \frac{|x_2-x_1|^{2\kappa}}{(t_2-s)^{\frac{2\kappa\beta}{\alpha}}} \int_{\mathbb{R}} \int_0^\infty u^{\frac{2\kappa\beta}{\alpha}} \left[p\left(\left((t_2-s)/u\right)^\beta, \frac{x_2-y}{2}\right) \right. \\
& \quad \left. + p\left(\left((t_2-s)/u\right)^\beta, \frac{x_1-y}{2}\right) \right]^2 g_\beta^2(u) du \rho(dy) \\
& \leq C \frac{|x_2-x_1|^{2\kappa}}{(t_2-s)^{\frac{2\kappa\beta}{\alpha}}} \int_0^\infty u^{\frac{2\kappa\beta}{\alpha}} g_\beta^2(u) \int_{\mathbb{R}} \left[p^2\left(\left((t_2-s)/u\right)^\beta, \frac{x_2-y}{2}\right) \right. \\
& \quad \left. + p^2\left(\left((t_2-s)/u\right)^\beta, \frac{x_1-y}{2}\right) \right] \rho(dy) du
\end{aligned} \tag{49}$$

In order to obtain the upper bound of the above quantity (49), we need to compute the inner integral

$$\int_{\mathbb{R}} p^2\left(\left((t_2-s)/u\right)^\beta, \frac{x_2-y}{2}\right) \rho(dy),$$

as follows by using (45).

$$\begin{aligned}
& \int_{\mathbb{R}} p^2\left(\left(\frac{t_2-s}{u}\right)^\beta, \frac{x_2-y}{2}\right) \rho(dy) \\
& \leq C \left(\frac{t_2-s}{u}\right)^{-\frac{2\beta}{\alpha}} \int_{\mathbb{R}} 1_{\left\{\left|\frac{x_2-y}{2}\right| \leq \left(\frac{t_2-s}{u}\right)^{\frac{\beta}{\alpha}}\right\}} \rho(dy) \\
& \quad + C \left(\frac{t_2-s}{u}\right)^{2\beta} \int_{\mathbb{R}} \frac{1}{\left|\frac{x_2-y}{2}\right|^{2(1+\alpha)}} 1_{\left\{\left|\frac{x_2-y}{2}\right| > \left(\frac{t_2-s}{u}\right)^{\frac{\beta}{\alpha}}\right\}} \rho(dy) \\
& \leq C \left[\max \left\{ \left(\frac{t_2-s}{u}\right)^{-\frac{\beta}{\alpha}}, \left(\frac{t_2-s}{u}\right)^{\frac{\beta(\bar{\delta}-2)}{\alpha}} \right\} + \left(\frac{t_2-s}{u}\right)^{2\beta - \frac{\beta(2\alpha+2-\bar{\delta})}{\alpha}} \right] \\
& \leq C \left[\left(\frac{t_2-s}{u}\right)^{-\frac{\beta}{\alpha}} + \left(\frac{t_2-s}{u}\right)^{\frac{\beta(\bar{\delta}-2)}{\alpha}} \right].
\end{aligned} \tag{50}$$

Similarly, one can also prove that

$$\int_{\mathbb{R}} p^2 \left(\left(\frac{t_2 - s}{u} \right)^{\beta}, \frac{x_1 - y}{2} \right) \rho(dy) \leq C \left[\left(\frac{t_2 - s}{u} \right)^{-\frac{\beta}{\alpha}} + \left(\frac{t_2 - s}{u} \right)^{\frac{\beta(\bar{\delta}-2)}{\alpha}} \right]. \quad (51)$$

Inserting the estimates (50) and (51) into (49), one gets that

$$\begin{aligned} & \int_{\mathbb{R}} |G_{t_2-s}(x_2 - y) - G_{t_2-s}(x_1 - y)|^2 \rho(dy) \\ & \leq C \frac{|x_2 - x_1|^{2\kappa}}{(t_2 - s)^{\frac{2\kappa\beta}{\alpha}}} \int_0^\infty u^{\frac{2\kappa\beta}{\alpha}} g_\beta^2(u) \left[\left(\frac{t_2 - s}{u} \right)^{-\frac{\beta}{\alpha}} + \left(\frac{t_2 - s}{u} \right)^{\frac{\beta(\bar{\delta}-2)}{\alpha}} \right] du. \end{aligned} \quad (52)$$

According the asymptotic behaviour of $g_\beta(u)$ as u tends to infinity, we have that

$$\int_0^\infty u^{\frac{(2\kappa+1)\beta}{\alpha}} g_\beta^2(u) du < \infty, \quad \text{and} \quad \int_0^\infty u^{\frac{2\kappa\beta+\beta(2-\bar{\delta})}{\alpha}} g_\beta^2(u) du < \infty,$$

if $0 < \kappa < \frac{2\alpha\beta+\alpha-2\beta+\beta\bar{\delta}}{\alpha}$. Based on this assumption on the positive constant κ , furthermore, we have that

$$\begin{aligned} & \int_0^{t_2} \int_{\mathbb{R}} |G_{t_2-s}(x_2 - y) - G_{t_2-s}(x_1 - y)|^2 \rho(dy) ds \\ & \leq C |x_2 - x_1|^{2\kappa} \int_0^{t_2} (t_2 - s)^{-\frac{2\kappa\beta}{\alpha}} \left[(t_2 - s)^{-\frac{\beta}{\alpha}} + (t_2 - s)^{\frac{\beta(\bar{\delta}-2)}{\alpha}} \right] ds, \end{aligned}$$

So if by choosing $0 < \kappa < \frac{\alpha+\beta(\bar{\delta}-2)}{2\beta}$, one has that

$$\int_0^{t_2} (t_2 - s)^{-\frac{2\kappa\beta}{\alpha}} \left[(t_2 - s)^{-\frac{\beta}{\alpha}} + (t_2 - s)^{\frac{\beta(\bar{\delta}-2)}{\alpha}} \right] ds < \infty.$$

Moreover, it is easy to check that $0 < \frac{\alpha+\beta(\bar{\delta}-2)}{2\beta} < \frac{2\alpha\beta+\alpha-2\beta+\beta\bar{\delta}}{\alpha}$. So we have that

$$\mathcal{N} \leq C |x_2 - x_1|^{2\kappa},$$

with $0 < \kappa < \frac{\alpha+\beta(\bar{\delta}-2)}{2\beta}$.

Thus we can conclude the proof of this lemma. ■

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